A Review of Deep Learning for Individualized Treatment Effect Estimation in Healthcare Time Series

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The proliferation of time series healthcare data presents a transformative opportunity for personalized healthcare. However, utilizing this observational data for clinical decision support requires moving beyond risk prediction to causal inference. Specifically, estimating time-varying individual treatment effects is crucial for answering "what-if" questions in dynamic healthcare settings, where patient states and treatment efficacy evolve continuously over time. While traditional statistical methods often struggle with the high dimensionality and non-linearity of modern healthcare data, deep learning has emerged as a powerful paradigm for modeling these complex temporal dependencies. This literature review synthesizes the theory behind time-varying ITE estimation and its application potential in clinical settings. We provide a comprehensive overview of the field, structured around theoretical foundations, deep learning-based methodologies, performance evaluation practices, and current clinical applications. Additionally, we discuss key unresolved challenges and outline a roadmap for future research.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**.

Additional Key Words and Phrases: Causal Inference, Individualised Treatment Effect, Time-Varying Treatment Effect, Deep Learning, Healthcare

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1 Introduction

The paradigm of modern healthcare is undergoing a profound transformation, shifting from generalized treatment protocols toward personalized healthcare. In clinical decision-making, one of the ultimate objectives is the ability to predict how a specific patient will respond to a specific treatment at a specific time. While traditional predictive modeling has excelled at estimating risk (e.g., predicting mortality given current vitals), it often fails to answer the critical counterfactual questions: "What would happen to this patient's blood pressure if we administered drug A instead of drug B right now?" Answering this requires moving beyond association to causal inference [76]. Unlike correlation, which merely observes that two events occur together, causality asserts that the intervention (the treatment) is the mechanism responsible for the outcome. Fundamentally, this re-frames the learning objective: rather than mapping observed features to observed labels, the goal is to predict outcomes under treatments that may not have occurred.

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Since a patient can only receive one treatment sequence at a time, the alternative outcomes—the *counterfactuals*—are never observed, effectively turning causal inference into a complex missing data problem [89].

This necessity for individualized prediction has motivated a transition from estimating the Average Treatment Effect (ATE) to the Individual Treatment Effect (ITE). While the former is a measure of how a treatment affects a population on average, the latter seeks to quantify the causal effect of an intervention at the level of a single patient unit. Clinical heterogeneity means that a treatment beneficial for the "average" patient may be ineffective or harmful for a specific individual. While early work in ITE estimation focused on static settings (a single decision made at a single point in time) [93], this view is misaligned with clinical reality. Healthcare is inherently dynamic and longitudinal. It is characterized by sequential decision-making, where treatments vary over time, patient states evolve in response to those treatments, and updated patient states influence future treatment decisions. This temporal process is conceptualized in Figure 1, which tracks the history of observed patient states (covariates) and observed treatments leading up to a specific decision time t. Consequently, time-varying ITE estimation has emerged as a critical topic of research, aiming to utilize this history to estimate the causal impact at a future instance, specifically illustrated as a τ-step-ahead prediction. As visually distinguished in the diagram, the core objective is to quantify the treatment effect by comparing the factual outcome—the observed value following the prescribed treatment sequence—against the counterfactual outcome, which represents the hypothetical value that would have been observed had an alternative treatment sequence been prescribed.

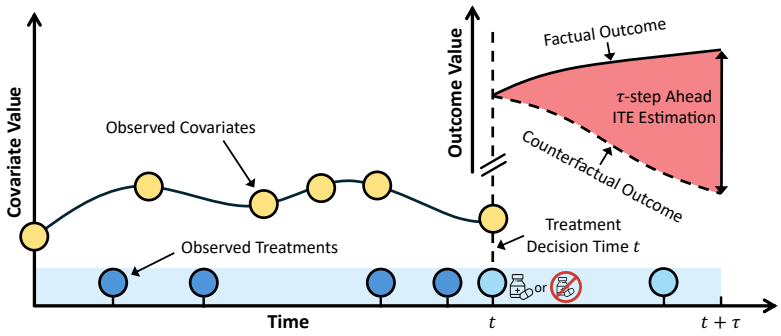

Fig. 1. Visualization of the τ-step-ahead time-varying ITE estimation.

The importance of time-varying ITE estimation is delineated by its diverse and profound applications that span the entire spectrum of care. In oncology, for instance, it can determine the optimal timing and sequence of chemotherapy and radiation to maximize survival while minimizing toxicity. In critical care settings like the Intensive Care Unit (ICU), it allows for the real-time adjustment of vasopressors and mechanical ventilation settings based on minute-by-minute physiological feedback. Similarly, in chronic disease management, it facilitates the personalization of medication schedules for conditions like diabetes, adapting to fluctuating biomarkers and lifestyle factors to improve long-term prognosis.

Realizing the full potential of these applications, however, relies heavily on data sources capable of capturing the complexity of longitudinal patient care. While Randomized Control Trials (RCTs) remain the gold standard for causal validation, they are practically ill-suited for training time-varying ITE estimation models; they are often constrained by high costs, small sample sizes, criticized for their ethics due to withholding care, and have rigid protocols that fail to reflect the heterogeneity of real-world populations [26, 37, 86]. Hence, estimation must rely on the abundance of

observational time series healthcare data. These datasets offer a rich, high-frequency historical record of patient states and treatments as they naturally evolve in clinical practice. However, relying on observational data introduces the fundamental challenge of time-dependent confounding. Unlike in RCTs, treatment assignment in observational data is non-random and biased by the patient's evolving health status; clinicians administer treatments based on symptoms that were themselves influenced by prior treatments. This creates a dangerous feedback loop: if an aggressive treatment is typically given only to severe patients, a standard model might erroneously learn that the aggressive treatment results in higher mortality. Disentangling the true causal effect of the treatment from the patient's underlying severity requires robust methodologies capable of correcting for this bias [38].

Historically, time-varying causal effects were estimated using statistical methods such as Marginal Structural Models [82] or G-computation [80]. Despite being theoretically sound, these traditional approaches struggle to process the high-dimensional, irregular, and non-linear data characteristic of modern time series healthcare data [58]. To address these limitations, the surge in deep learning approaches in recent years has been applied to causal inference. Neural network architectures such as Recurrent Neural Network (RNN) [9, 58] and Transformer [65] have demonstrated a superior ability to learn latent representations of patient history, model complex non-linear interactions between covariates and treatments, and handle the long-range dependencies inherent in patient history. These advancements represent a significant leap forward, enabling more reliable time-varying ITE estimations from time series healthcare data.

Table 1. Comparison of recent surveys and reviews on ITE estimation. The table indicates whether each work addresses time-varying settings (TV), accounts for deep learning methods (DL), causal inference and ITE theory, clinical applications, framework evaluation, and an approximate time-varying ITE literature count.

Review	Year	TV	DL	Theory	Application	Evaluation	#Literature
Bica et al. [10]	2021	✓	✓	✓	✓	✓	5+
Shi et al. [94]	2022			✓	✓		
Feuerriegel et al. [24]	2024		✓	✓	✓	✓	
Zhang et al. [113]	2024	✓		✓			5+
Munroe et al. [68]	2025				✓		
Ghosheh et al. [30]	2025	✓	✓			✓	5+
Our Work	2026	✓	✓	✓	✓	✓	40+

Despite the rapid growth of the causal inference literature, current reviews remain fragmented. As illustrated in Table 1, existing surveys typically prioritize isolated aspects, such as theoretical understanding or methodological categorization, but fail to provide a unified analysis of the use of deep learning for time-varying ITE estimation in healthcare. In particular, recent surveys such as [24, 68, 94] focus on the static setting and do not address the temporal complexities inherent in time series patient data. Other reviews, such as [94, 113], discuss theoretical aspects but neglect recent advancements in deep learning frameworks. Even among reviews that discuss time-varying methods [10, 30, 113], the scope of coverage is often narrow. Our work distinguishes itself by comprehensively summarizing existing methods, significantly exceeding the breadth of previous reviews to provide an updated summary of the landscape. To our knowledge, this work is the first to bridge this gap by offering a comprehensive guide that spans theoretical foundations, deep learning methodologies, clinical applications, and model evaluation. Specifically, our primary contributions are detailed as follows:

(1) Theoretical Foundation: We provide a detailed and intuitive explanation of causal inference concepts, such as identifiability assumptions and time-dependent confounder adjustment, which are essential for unbiased time-varying ITE estimation.
(2) Methodological Taxonomy: We perform a comprehensive survey of deep learning frameworks for time-varying ITE estimation in healthcare, proposing a taxonomy structured around the **locus of adjustment**. By categorizing methods based on where the adjustment is applied—the input space, model structure, latent representation, or loss function—we go beyond architectural comparisons to distill the theoretical basis that justifies the ability of each mechanism to mitigate time-dependent confounding.
(3) Performance Evaluation: We analyze the critical components of framework validation, including the generation and use of synthetic versus real-world datasets, the selection of appropriate performance metrics, and the summarization of baselines that are predominantly used for comparative analysis.
(4) Future Roadmap: We identify and discuss critical open problems and emerging research avenues, providing a roadmap for future work in areas such as time-to-event outcome modeling, multi-modal data integration, and the development of medical digital twins.

Our review is organized to systematically guide the reader through the landscape of time-varying ITE estimation, bridging theoretical underpinnings with practical implementation. We begin in Section 2 by outlining the search strategy employed to curate the relevant literature. Section 3 establishes the necessary theoretical foundation, defining foundational concepts, illustrating the specific challenge of time-dependent confounding, and detailing the identifiability assumptions required for valid causal effect estimation. Transitioning from abstract theory to technical execution, Section 4 delineates the standard training and inference setups. The analysis of methodologies appears in Section 5, which presents a comprehensive taxonomy of methodologies categorized by their adjustment strategies, followed by an examination of their real-world clinical applications in Section 6. Section 7 addresses the critical issue of model validation, examining the datasets, metrics, and baselines essential for assessing performance in the absence of ground truth. We conclude by analyzing current challenges in the field, such as assumption violations and data challenges, in Section 8, and proposing future research directions that incorporate time-to-event outcomes, multi-modal integration, and the development of digital twins in Section 9.

2 Search Strategy

We conducted a targeted literature search using Google Scholar to identify methods for ITE estimation from time series health data. Our search keywords combined terms such as *time-varying*, *individualized treatment effect*, *conditional average treatment effect*, *heterogeneous treatment effect*, and *time-varying confounders*. In addition to keyword-based searches, we performed a recursive citation search starting from influential early works in the field, including R-MSN [58], CRN [9], G-Net [56], CT [65], Time Series Deconfounder [8], DTA [54], and Sequential Deconfounder [35].

3 Theoretical Foundation

This section explains the theoretical foundation for time-varying ITE estimation in time series health data. To begin with, we introduce the core concepts necessary to define the problem, including a categorization of time-varying ITE settings, the notation, and the target estimand. We then illustrate the problem of time-dependent confounding and the need for appropriate adjustment. Finally, we present the sequential assumptions that guarantee the identifiability of the causal estimand and describe how identification and estimation proceed to compute the time-varying ITE.

3.1 Foundational Concepts

A central goal of causal inference is to move beyond statistical associations and estimate the causal effect of a treatment or intervention on an outcome of interest (We will use the terms *treatment* and *intervention* interchangeably). This objective is characterized by answering counterfactual questions, such as, "What would have been the outcome for this patient had they received a different medication?" in a static setting or, in a time-varying context, "What would have been this patient's 30-day survival had we followed a different sequence of vasopressor dosages over their ICU stay?" Time-varying ITE estimation subsequently seeks to quantify how such causal effects vary across patients, given their medical histories.

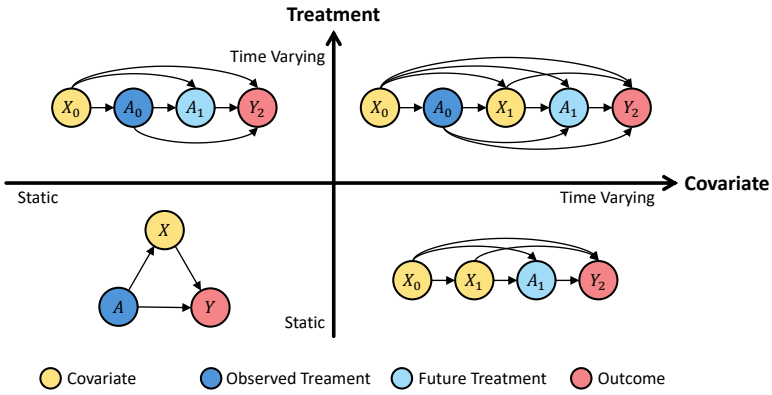

Fig. 2. Categorization of ITE into 4 quadrants: (1) static covariates & static treatments; (2) time-varying covariates & static treatments; (3) static covariates & time-varying treatments; and (4) time-varying covariates & time-varying treatments.

Problems in ITE estimation can be broadly categorized by the time-dependent nature of the available data. At a high level, this data consists of covariates (features describing the patient state, such as age or vital signs) and treatments (interventions applied to the patient, such as drugs or medical procedures). Even though the formal mathematical definitions for these variables will be presented in the ensuing paragraphs, it is useful to first categorize estimation problems based on whether these elements are static (fixed at a single point) or time-varying (evolving over time). This distinction creates four distinct quadrants of analysis (Figure 2):

(1) **Static Covariate & Static Treatment**: The classic causal inference problem in which baseline (pre-treatment) covariates are used to estimate the effect of a single fixed treatment. This is the fundamental scenario for answering questions like, "Does Drug A work better than a placebo for patients with a given baseline disease severity?"

(2) **Time-Varying Covariate & Static Treatment**: A single treatment is administered, but the decision to treat (or not) is based on evolving covariates. This setting is common in clinical scenarios where clinicians use longitudinal health records to recommend a one-time intervention. For instance, deciding whether and when to perform an organ transplant for a patient on a waiting list. The decision is based on their evolving weekly lab results and health vitals. After the transplant (or non-transplant), the patient's long-term survival is measured. This differs from setting (4) by the absence of a feedback loop where treatment alters the future covariate trajectory for subsequent treatment decisions.

(3) **Static Covariate & Time-Varying Treatment**: A sequence of treatments is administered, but the entire treatment plan is based solely on baseline static covariates. While uncommon in healthcare, this setting is more plausible in industry. For example, a material is processed according to a pre-specified process "recipe" that depends only on baseline measurements. Initially, engineers observe static covariates describing the material (e.g., initial composition, moisture content, particle size, etc.). Based solely on these characteristics, they select a fixed sequence of control actions, such as temperature and pressure set points, at each minute of a furnace run. This control trajectory is then executed automatically and does not adapt to any intermediate sensor readings during processing. The final outcome (e.g., yield or product quality) thus depends on a time-varying treatment sequence that is entirely determined by static baseline covariates.

(4) **Time-Varying Covariate & Time-Varying Treatment**: Both covariates and treatments are subject to change and may influence one another over time. This is the most common and complex scenario, such as managing a patient in the ICU. An example would be when a physician repeatedly measures vital signs, uses them to administer a dose of a blood pressure drug, and then observes the new vital signs to decide on the next dose, aiming to optimize a downstream outcome such as 30-day survival.

This review focuses on the complexities introduced by time series health data. We will therefore set aside the standard static-static case (1). On top of that, the setting (3) with static covariates and time-varying treatments is less representative of routine clinical decision-making. This is because it implies that doctors make sequential treatment decisions without considering the patient's evolving health state. After each treatment, we expect the patient covariates to change in response, which should subsequently be accounted for in the decision-making that follows. Hence, our focus is on the remaining two settings (2 and 4), which together constitute the domain of time-varying ITE estimation.

To formalize time-varying ITE estimation, we adopt the Potential Outcomes (PO) framework (also known as the Neyman-Rubin Causal Model), originally developed for static treatment effect estimation [47, 71, 87], and extended to the time-varying setting [38, 80]. The PO framework provides the formal tools to define, identify, and estimate time-varying treatment effects. We first introduce the notation for the most general setting (4), from which the other cases can be viewed as special cases. All notations are summarized in Appendix A.

Throughout, we adopt the following conventions: (i) Capital letters without an overbar (e.g. $X_k,\ A_k,\ H_k$) denote random variables or random vectors. (ii) Lowercase letters (e.g. $x_k,\ a_k,\ h_k$) denote fixed values in the support of the corresponding random variables or random vectors. These may represent observed histories (realized data) or prescribed/hypothetical treatment sequences used to index potential outcomes. (iii) Calligraphic symbols (e.g. $\mathcal{A},\ \mathcal{H}_k$) denote non-random sets or spaces. Consider a longitudinal study where our goal is to estimate a causal effect at a decision time t on an outcome that will be measured τ time steps in the future, at time $t+\tau$. At each discrete time step $k \in \{0, \ldots, t, \ldots\}$, we observe a sequence of data for a single individual (we omit the individual index i for notational simplicity in most cases):

- (Pre-treatment) Covariates $X_k \in \mathbb{R}^{d_x}$: A d_x-dimensional random vector of patient features at time k. These represent the patient's state and can include both time-varying features (e.g., vital signs) and static features (e.g., demographics). A particular realized covariate vector is denoted $x_k \in \mathbb{R}^{d_x}$.
- Treatment A_k: The treatment assignment at time k is a random vector $A_k = [A_{k,1},\ \ldots,\ A_{k,d_a}]^\top$ encompassing d_a concurrent treatment options. A_k takes values in a joint treatment space $\mathcal{A}$, defined as the Cartesian product of the individual component domains $\mathcal{A} = \prod_{j=1}^{d_a} \mathcal{A}_j$, where $\mathcal{A}_j$ denotes the set of admissible dosages for the j-th

treatment type. This accommodates mixed treatment types: for a *discrete* treatment j, the component domain is a finite set of predetermined dosage levels $\mathcal{A}_j = \{\alpha_{j,1}, \ldots, \alpha_{j,m_j}\}$, where m_j is the number of available levels and α_j represents the predetermined scalar values; for a *continuous* treatment j, the component domain is a non-negative real value $\mathcal{A}_j = [0, \infty)$. A realized treatment vector at time k is denoted by $a_k = [a_{k,1}, \ldots, a_{k,d_a}]^\top \in \mathcal{A}$.

We use an overbar to denote the sequence of a variable up to a given time, for example,

$$\overline{X}_t = (X_0, \ldots, X_t), \qquad \overline{A}_t = (A_0, \ldots, A_t).$$

At decision time t, before the treatment A_t is chosen, the *observed history* is a random vector

$$H_t = (\overline{X}_t, \overline{A}_{t-1}),$$

which consists of all covariates up to and including X_t and all past treatments up to and including A_{t-1}. While H_t represents the random vector of history, $h_t = (\overline{x}_t, \overline{a}_{t-1}) \in \mathcal{H}_t$ is a particular realization of the history. Our outcome of interest is $Y_{t+\tau}$, measured τ steps after the decision time t. This outcome may be binary $Y_{t+\tau} \in \{0, 1\}$ (e.g., mortality) or continuous $Y_{t+\tau} \in \mathbb{R}$ (e.g., blood pressure). The static cases arise as special instances of this notation. In setting (2), with time-varying covariates and a single static treatment administered at decision time t, the treatment history $\overline{A}_{t-1}$ consists of a sequence of null (control) interventions, reflecting that no intervention has occurred prior to the current decision.

We are interested in the effect of the treatment sequence from t until the outcome is measured; more specifically, on the sequence of treatment vectors $(A_t, \ldots, A_{t+\tau-1})$. For notational simplicity, we define causal effects in this section by comparing *static* treatment sequences in the future. A static sequence of treatments from t to $t + \tau - 1$ is a fixed, non-random sequence of vectors defined as:

$$\overline{a}_{[t,\ t+\tau-1]} := (a_t, a_{t+1}, \ldots, a_{t+\tau-1}) \in \mathcal{A}^\tau,$$

where each $a_k \in \mathcal{A}$ is a specified treatment. It should be emphasized that $\overline{a}_{[t,t+\tau-1]}$ indexes a prescribed sequence of treatment values; it is not a realized sequence. In the potential outcomes framework, the outcome at time $t + \tau$ is conceptualized as a function of the future treatment sequence conditioned on the patient history. Let

$$Y_{t+\tau}\left(\overline{a}_{[t,\ t+\tau-1]}\right)$$

denote the potential outcome that would be observed at time $t + \tau$ if the future treatments from t to $t + \tau - 1$ were set to the fixed sequence $\overline{a}_{[t,t+\tau-1]}$. For brevity, we write the potential outcomes as $Y^{\overline{a}}_{t+\tau}$ and $Y^{\overline{a}'}_{t+\tau}$, where the superscripts $\overline{a} = \overline{a}_{[t,t+\tau-1]}$ and $\overline{a}' = \overline{a}'_{[t,t+\tau-1]}$ are prescribed with different future treatment sequences. It is noteworthy that *dynamic*, history-dependent treatment rules (dynamic treatment regimes) can be represented in the same framework by replacing static treatment sequences with mappings from histories to treatments [38, 69].

The *causal estimand* is the theoretical (typically unobserved) quantity that encodes the causal effect of interest. An *estimator* is a statistical procedure applied to finite data to approximate that estimand. The most granular estimand is the τ-step-ahead ITE:

Definition 3.1 (Time-varying Individualized Treatment Effect (ITE)). Fix a decision time t, a horizon $\tau \geq 1$, and two static future treatment sequences $\overline{a}$ and $\overline{a}'$ over time points $\{t,\ t + \tau - 1\}$. The τ-step-ahead time-varying ITE for individual i is

$$\mathrm{ITE}_{i,t,\tau}(\overline{a}, \overline{a}') = Y^{\overline{a}}_{i,\ t+\tau} - Y^{\overline{a}'}_{i,\ t+\tau}. \tag{1}$$

We re-introduce index i here to emphasize that the ITE is a quantity specific to a single unit, capturing the heterogeneity of treatment effects across individuals that is otherwise lost in population-level averages.

However, identifying the ITE is impossible due to the *fundamental problem of causal inference* [45]: for any individual i at time $t + \tau$, we observe only the factual outcome $Y_{i,t+\tau}$ corresponding to the treatment trajectory that the patient actually received. The alternative outcome is a missing *counterfactual*. This is intuitive, as one cannot simultaneously "prescribe" and "not prescribe" a medication for an individual. Because the ITE is unobservable, the target must be relaxed to estimable quantities that average over populations or subpopulations. The time-varying **Average Treatment Effect (ATE)** compares two treatment sequences at the population level:

$$\mathrm{ATE}_{t,\tau}(\overline{a},\ \overline{a}') = \mathbb{E}\left[Y^{\overline{a}}_{t+\tau} - Y^{\overline{a}'}_{t+\tau}\right]. \tag{2}$$

While this provides a useful population-level summary, the ATE does not capture treatment effect heterogeneity, which is central to personalized healthcare. A more informative target is the time-varying **Conditional Average Treatment Effect (CATE)**, which conditions on an individual's history.

Definition 3.2 (Time-varying Conditional Average Treatment Effect (CATE)). The τ-step-ahead time-varying CATE at decision time t is the conditional expectation of the difference between potential outcomes given the observed history H_t.

$$\delta_{t,\tau}(h_t;\ \overline{a},\ \overline{a}') = \mathbb{E}\left[Y^{\overline{a}}_{t+\tau} - Y^{\overline{a}'}_{t+\tau}\,\middle|\,H_t = h_t\right], \tag{3}$$

Formally, $\delta_{t,\tau}(h_t;\ \overline{a},\ \overline{a}')$ is a conditional average effect rather than an individual-level quantity. However, as the conditioning h_t becomes sufficiently rich and high-dimensional, the conditional average can serve as a practically meaningful proxy for the individualized effect $\mathrm{ITE}_{i,t,\tau}$, especially in the context of individualized decision support. This perspective underlies much of the literature on ITE estimation in both static and time-varying settings [9, 93], where the estimand in Equation (3) is often informally referred to as the time-varying ITE. We adopt this convention throughout, and when the treatment sequences $(\overline{a}, \overline{a}')$ and times (t, τ) are fixed or clear from the context, we write $\delta(h_t)$ for brevity.

3.2 The Challenge of Time-Dependent Confounding

While the time-varying CATE is our target estimand, we must now ask how to estimate it from the data. It is crucial to distinguish between two data-gathering settings: RCTs and observational studies. While some sequential trial designs exist [55, 70], the vast majority of time series health data originate from observational sources, such as Electronic Health Records (EHRs). In an ideal RCT, the random assignment of treatment indicates that the treatment and control groups are, on average, identical in their characteristics. In observational data, this statement is not true. Treatment selection is determined by clinicians or patients, and these decisions depend on the evolving patient state over time: current covariates as well as past covariates and treatments. These temporal dependencies give rise to **time-dependent confounding**, where a covariate X_k acts as a common cause for both the immediate treatment assignment A_k and the final outcome $Y_{t+\tau}$. Because such covariates simultaneously drive treatment decisions and future outcomes, patients receiving different treatment sequences can differ systematically in their prognoses even before the treatment choice is made, inducing confounding bias in naive comparisons of treatment effects over time. It is important to distinguish this from selection bias, a distinct causal phenomenon often conflated in the literature. Structurally, *confounding bias* arises from the existence of a backdoor path through common causes. In contrast, *selection bias* arises from conditioning on a common effect of the treatment and outcome—structurally known as a *collider* [38].

Consider a heart medication prescribed over multiple clinic visits. At each visit, the clinician observes the patient's current disease severity (e.g., symptoms, blood pressure, ejection fraction) and uses this evolving state to decide whether to start, stop, or adjust the medication. Disease severity at each visit is then a time-varying confounder: it is affected by earlier treatment decisions, helps to determine the next treatment choice, and remains predictive of future outcomes, such as hospitalization or death. Consequently, patients who follow more "aggressive" treatment sequences over time are typically those whose disease is worse or deteriorating. A naive comparison of outcomes between aggressively treated patients and those receiving milder or no treatment will, therefore, tend to conclude that aggressive treatment is ineffective or even harmful when, in fact, much of the worse prognosis is driven by the underlying, time-evolving severity rather than by the medication itself.

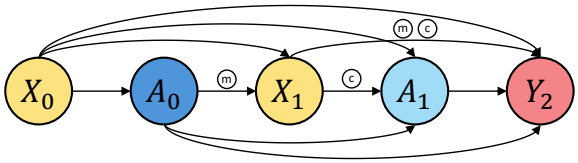

Fig. 3. Causal graph of 2-step time-varying covariates $(X_0,\ X_1)$ and time-varying treatments $(A_0,\ A_1)$ resulting in the outcome Y_2. ⓜ and ⓒ denote the mediator and confounder paths, respectively.

We illustrate these causal relationships using a directed acyclic graph (DAG). Figure 3 provides an example of a 2-step time-varying treatment effect estimation setting with nodes X_0, A_0, X_1, A_1, and Y_2. It can be seen that the covariate X_k (e.g., X_1) is a cause of both the subsequent treatment A_k (e.g., A_1) and the outcome Y_2, thus acting as a time-varying confounder. On top of that, this same covariate X_k (e.g., X_1) is also caused by past treatment A_{k-1} (e.g., A_0) and lies on the causal path to the outcome Y_2. This creates a critical analytical dilemma by giving X_k a dual role:

- It is a *mediator*: the confounder is caused by past treatment ($A_{k-1} \to X_k$) and also affects the outcome ($X_k \to Y$), forming a mediating path ($A_{k-1} \to X_k \to Y$). This path is a valid part of the total causal effect of past treatment.
- It is a *confounder*: confounders affect future treatment ($X_k \to A_k$) while influencing the outcome ($X_k \to Y$); therefore, it is a common cause of A_k and Y. This creates a spurious, non-causal backdoor path ($A_k \leftarrow X_k \to Y$).

In order to estimate the causal effect of the treatment sequence $\overline{a}$, we must block the confounding path ($A_k \leftarrow X_k \to Y$) at each step k. However, if we condition on X_k using standard methods (e.g., regression), we are conditioning on a mediator, which blocks the valid causal path ($A_{k-1} \to X_k \to Y$). Therefore, a valid adjustment procedure is required.

Definition 3.3 (Adjustment). In the time-varying setting, adjustment is a sequential procedure that neutralizes the confounding effect of variables X_k at each step k, without blocking the causal effects of past treatments $A_{j<k}$ that are mediated through those same variables.

Standard adjustment methods, such as multiple regression [4] or static propensity score matching [85], fail in this setting. These methods cannot distinguish between X_k's dual roles; they condition on the entire history, thereby blocking the confounding path but also incorrectly blocking the valid causal mediation path [38, 80]. This failure necessitates a different and more sophisticated set of assumptions, which we introduce next to provide identifiability for the ITE estimand, rendering the causal effect recoverable from observational data.

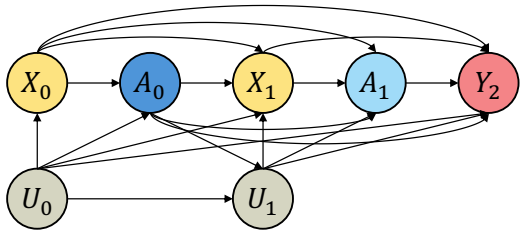

Fig. 4. Causal graph of 2-step time-varying covariates $(X_0,\ X_1)$ and time-varying treatments $(A_0,\ A_1)$ resulting in the outcome Y_2 with the presence of unmeasured confounders $(U_0,\ U_1)$.

3.3 From Identification to Estimation

To address confounding and obtain unbiased estimates of the time-varying ITE, we must impose assumptions on the data generating process (DGP). These assumptions are used to establish *identifiability*: the property that a causal estimand can, in principle, be expressed as a functional of the observed data distribution. Recall that the estimand of interest at decision time t and horizon τ is the time-varying ITE in Equation (3):

$$\delta(h_t) = \mathbb{E}\left[Y^{\overline{a}}_{t+\tau} - Y^{\overline{a}'}_{t+\tau} \,\middle|\, H_t = h_t\right].$$

Because earlier treatments affect subsequent covariates, which in turn affect both future treatment decisions and outcomes, we require assumptions that are formulated sequentially over time.

(1) **Consistency.** Let $\overline{a}_{[0,t+\tau-1]} = (\overline{a}_{t-1}, \overline{a}_{[t,t+\tau-1]})$ denote a fixed full treatment sequence, where $\overline{a}_{t-1}$ is a particular realized past treatment sequence (a value of $\overline{A}_{t-1}$) and $\overline{a}_{[t,t+\tau-1]}$ is a fixed future treatment sequence. If, for an individual, the observed treatment sequence matches $\overline{a}_{[0,t+\tau-1]}$, i.e.

$$\overline{A}_{t-1} = \overline{a}_{t-1} \quad \text{and} \quad \overline{A}_{[t,t+\tau-1]} = \overline{a}_{[t,t+\tau-1]},$$

then the observed outcome equals the corresponding potential outcome:

$$Y_{t+\tau} = Y_{t+\tau}(\overline{a}_{[0,\ t+\tau-1]}).$$

This implies that there are no multiple versions of the treatment. For example, if A_k = surgery but surgical skill varies across surgeons in ways that affect the outcome, then $Y(\text{surgery})$ is ill-defined, as different surgeons implicitly correspond to different interventions. Consistency requires that the treatment definition be sufficiently precise so that technical variations do not change the potential outcome. It is noteworthy that in the static ITE estimation setting, the assumption required is the **Stable Unit Treatment Value Assumption (SUTVA)** consisting of 2 components: (i) *consistency*, as defined above, and (ii) *no interference*, stating that an individual's potential outcomes do not depend on the treatment assignments of others. In modern deep learning literature, the "No Interference" component is frequently subsumed into the statistical setup. Seminal works [9, 56] explicitly list "Consistency" as an assumption but encode "No Interference" implicitly by defining the dataset as a collection of N independent and identically distributed (i.i.d.) patient trajectories.

(2) **Sequential Positivity**. For time-varying treatments that may be either discrete or continuous, sequential positivity is naturally stated in terms of the conditional distribution of A_k given the history H_k. For each time step $k \in \{t,\ \ldots,\ t+\tau-1\}$ and each history $h_k \in \mathcal{H}_k$ with $\Pr(H_k = h_k) > 0$, the conditional distribution of A_k given H_k must assign positive mass or density to every treatment value a_k that appears in a treatment sequence

of interest. Concretely, if A_k is discrete, then for all realized histories h_k, and all a_k that are components of $\overline{a}$ or $\overline{a}'$:

$$\Pr(A_k = a_k \mid H_k = h_k) > 0,$$

whereas if A_k is continuous with conditional density $f_{A_k|H_k}$, we require

$$f_{A_k|H_k}(a_k \mid h_k) > 0,$$

More intuitively, the effect of a given treatment sequence cannot be identified for histories where that sequence (or parts of it) is impossible under the observed data. If a particular treatment a_k is never administered to patients with a history $H_k = h_k$, then we cannot use the observed data to learn about the counterfactual outcome under that action in that stratum.

(3) **Sequential Exchangeability (or Ignorability)**. While often termed "Sequential Strong Exchangeability" to imply the joint satisfaction of exchangeability and positivity [85], we list sequential positivity separately above for clarity. At each time step from the baseline to the end of the treatment horizon $k \in \{0, \ldots, t+\tau-1\}$, after conditioning on the history H_k, the treatment A_k is independent of the future potential outcomes under *any* treatment sequence. For the horizon-τ outcome, this can be written as

$$Y_{t+\tau}(\overline{a}_{[0,\ t+\tau-1]}) \perp\!\!\!\perp A_k \mid H_k \quad \forall k \in \{0, \ldots, t+\tau-1\},\ \forall \overline{a}_{[0,\ t+\tau-1]} \in \mathcal{A}^{t+\tau}.$$

In practice, treatment choices A_k often depend on disease severity, trends in biomarkers, and other information encoded in the patient history H_k. Sequential exchangeability asserts that conditioning on H_k is sufficient to block all backdoor paths and that there are no additional unmeasured confounders that jointly affect treatment and future outcomes. Equivalently, treatment assignment is "as good as randomized" within each stratum of H_k. It is important that sequential exchangeability applies to the entire history $k \in \{0, \ldots, t+\tau-1\}$, not just to the future treatment time window $k \in \{t, \ldots, t+\tau-1\}$. This is because if exchangeability is violated in the past, the observed history H_t may fail to capture hidden confounders that influence both past treatments and future outcomes, rendering H_t a proxy for unmeasured variables rather than a sufficient summary of the causal state, and the eventual estimation will consists of bias from the unadjusted past confounders.

The sequential exchangeability assumption fails if latent variables U_k that affect both treatment and outcome are missing from the patient history H_k. To make this concrete, consider the DAG in Figure 4, which has decision time $k = 1$ with measured confounders $X_0,\ X_1$, treatments $A_0,\ A_1$, and an outcome at time $k = 2$, Y_2. At time $k = 1$, the observed history is H_1 (e.g., $H_1 = (X_0, A_0, X_1)$), while an unmeasured confounder U_1 influences both A_1 and $Y_2^{\overline{a}}$. Because $U_1 \notin H_1$, the backdoor path

$$(A_1 \leftarrow U_1 \rightarrow Y_2^{\overline{a}})$$

remains open after conditioning on H_1. Formally, since

$$\Pr(A_1 \mid H_1,\ U_1) \neq \Pr(A_1 \mid H_1)$$

for some treatment value $a_1 \in \mathcal{A}$, and the potential outcome $Y_2^{\overline{a}}$ depends on U_1, it follows that

$$Y_2^{\overline{a}} \not\perp\!\!\!\perp A_1 \mid H_1.$$

Intuitively, the unmeasured confounder U_1 drives both the clinician's treatment decision A_1 and the future outcome $Y_2^{\overline{a}}$; even within strata of H_1, treatment remains informative about counterfactual outcomes, thereby violating sequential

exchangeability. The violation shown in this simplified setup at time $k = 1$ is a special case of a general problem that can occur at any time step k in a time-varying setting. Whenever an unmeasured variable U_k exists that is not included in the observed history $U_k \notin H_k$ and that affects both the treatment A_k and the future potential outcome $Y^{\bar{a}}_{t+\tau}$, the path

$$A_k \leftarrow U_k \rightarrow Y^{\bar{a}}_{t+\tau}$$

remains open after conditioning on H_t, and the violation persists as

$$Y^{\bar{a}}_{t+\tau} \not\perp\!\!\!\perp A_k \mid H_k.$$

Under consistency, sequential positivity, and sequential exchangeability, the time-varying ITE estimand $\mathbb{E}[Y^{\bar{a}}_{t+\tau} \mid H_t = h_t]$ is formally identified. It is useful to distinguish between identifiability, identification, and estimation. While *identifiability* is a property granted by assumptions on the DGP, *identification* explicitly expresses the target estimand as a functional of observed data distributions. Provided with the identification formula, *estimation* is performed to approximate the unknown distributions in the functional. In practice, the extent to which identification is made explicit and distinct from estimation varies across approaches. Some methods state the identification formula explicitly by leveraging the G-formula [56] or the Inverse Probability of Treatment Weighting (IPTW) [58]. In these cases, adjustment for time-varying confounders is encoded in the identification formula using standardization [38, 80] or population reweighting [38, 82]. Other approaches maintain the same assumptions and estimand but bypass explicit analytical summation. They leverage the flexibility of neural networks to directly estimate the identified functional $h_t \mapsto \mathbb{E}[Y^{\bar{a}}_{t+\tau} \mid H_t = h_t]$ [9, 65]. Such approaches perform adjustment directly in the framework design during estimation, blurring the boundary between identification and estimation.

Table 2. Taxonomy of ITE estimation methodologies for time series health data. For each method, we present the name, publication year, ITE Category (corresponding to the quadrants defined in Figure 2), outcome and treatment types (continuous and/or discrete), specific adjustment strategies employed, and open-source availability.

Name	Year	ITE Category	Outcome[a]	Treatment[a]	Adjustment	Open-sourced
SyncTwin [78]	2021	(3)	Continuous	Discrete	M	✓
SCouT [20]	2023	(3)	Continuous	Discrete	M	✓
T4 [61]	2024	(4)	Continuous	Discrete	M	✓
G-Net [56]	2020	(4)	Both	Continuous	GF	
Su et al. [98]	2024	(4)	Both	Continuous	GF	
G-Transformer [109]	2024	(4)	Both	Both	GF	
Deng et al. [21]	2024	(4)	Continuous	Both	GF	
G-Transformer [41]	2024	(4)	Continuous	Discrete	GF	✓
CURE [60]	2024	(3)	Discrete	Discrete	GF	✓
R-MSN [58]	2018	(4)	Continuous	Discrete	IPTW	✓
Wu et al. [107]	2024	(4)	Continuous	Discrete	IPTW	✓
SCIP-Net [40]	2025	(4)	Continuous	Discrete	IPTW	✓
IVW-DR-learner [25]	2025	(4)	Continuous	Discrete	DR	✓
CRN [9]	2020	(4)	Continuous	Discrete	RB	✓
CT [65]	2022	(4)	Continuous	Discrete	RB	✓
TE-CDE [91]	2022	(4)	Continuous	Discrete	RB	✓
TCFimt [108]	2022	(4)	Continuous	Discrete	RB	
CF-GODE [48]	2023	(4)	Continuous	Discrete	RB	
RCAB [115]	2023	(4)	Continuous	Continuous	RB	
CAG-ODE [46]	2024	(4)	Continuous	Discrete	RB	✓
ACTIN [23]	2024	(4)	Continuous	Both	RB	✓
Causal CPC [23]	2024	(4)	Continuous	Discrete	RB	
OptAB [106]	2024	(4)	Continuous	Discrete	RB	✓
CEET [111]	2025	(4)	Continuous	Discrete	RB	
Liu et al. [63]	2025	(4)	Continuous	Discrete	RB	
EDTS [72]	2025	(4)	Continuous	Discrete	RB	✓
Mamba-CDSP [103]	2025	(4)	Continuous	Discrete	RB	
Guski et al. [33]	2025	(4)	Continuous	Discrete	RB	✓
TV-SurvCaus [1]	2025	(4)	Continuous	Continuous	RB, IPTW	
DCRN [7]	2021	(4)	Continuous	Discrete	RD	
DR-CRN [17]	2022	(4)	Continuous	Discrete	RD	✓
CDS [114]	2021	(4)	Continuous	Discrete	-	✓

Continued on next page...

Name	Year	ITE Category	Outcome[a]	Treatment[a]	Adjustment	Open-sourced
Kim et al. [52]	2024	(4)	Continuous	Continuous	-	
BNCDE [43]	2024	(4)	Continuous	Discrete	N/A	✓
COSTAR [66]	2024	(4)	Continuous	Both	O	✓
INSITE [51]	2024	(4)	Continuous	Both	O	✓
TSD [8]	2020	(4)	Continuous	Discrete	N/A	✓
DSW [62]	2020	(4)	Continuous	Discrete	IPTW	✓
DTA [54]	2021	(4)	Continuous	Discrete	N/A	✓
LipCDE [12]	2023	(4)	Continuous	Discrete	IPTW	
LipSCDE [11]	2023	(4)	Continuous	-	IPTW	
VTDNet [18]	2025	(4)	Continuous	Discrete	IPTW	✓
CF-ODE [19]	2022	(3)	Continuous	Discrete	GF	✓

Framework Abbreviations: TSD = Time Series Deconfounder; SD = Sequential Deconfounder.
Treatment Abbreviations: "-" = unknown.
Adjustment Abbreviations: GF = G-Formula; IPTW = inverse probability of treatment weighting; DR = doubly robust; RB = representation balancing; RD = representation disentanglement; M = matching; O = other methods; "N/A" = not applicable; "-" = does not perform proper adjustment.

[a] Classifications for outcome and treatment types (continuous and/or discrete) reflect the specific settings used in the experimental evaluation, rather than the theoretical capabilities of the underlying model architectures.

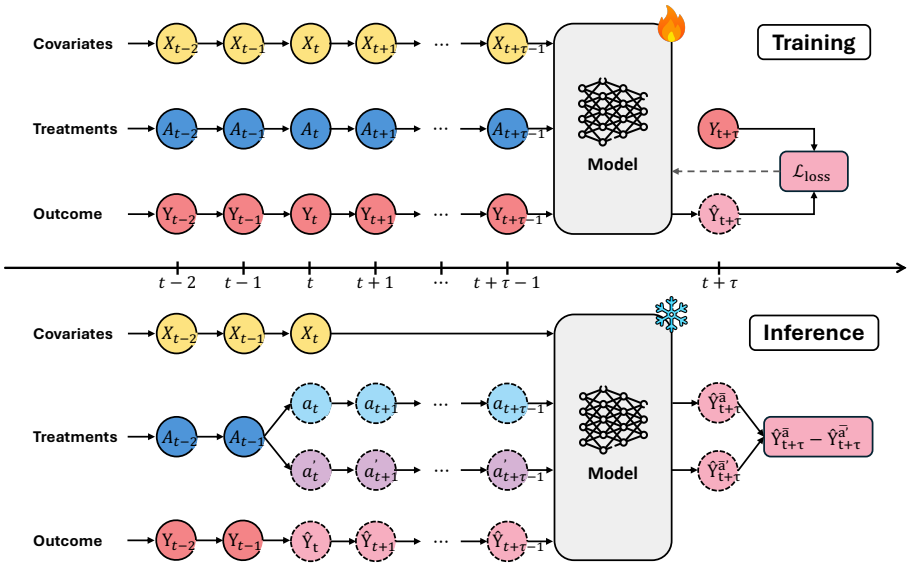

Fig. 5. Visualization of the model training and inference stages for time-varying ITE estimation. The 🔥 and ❄ icon separately denotes trainable and frozen model parameters. Solid and dashed circular outlines distinguishes between observed data and predicted/prescribed values.

4 From Theory to Practice

Now that the theoretical foundation for estimating the ITE from time series health data has been established, we provide a concise outline of the procedure for using observational patient data for training and inference. This bridges the gap between theory and framework implementation and provides the necessary intuition to develop a better understanding of the ensuing ITE estimation methodologies.

The observational dataset consists of time-varying covariates $x_{i,k}$, treatments $a_{i,k}$, and treatment effects/outcomes $y_{i,k+1}$ for patient i, which are denoted as

$$\mathcal{D} = \left\{ \left(x_{i,k},\ a_{i,k},\ y_{i,k+1} \right)_{k=0}^{T_i} \right\}_{i=1}^{N}$$

for a total of N i.i.d. patients, each with a sequence length T_i. Note that the time index for the outcome shifts to $k+1$ because the effect of a treatment prescribed at time k manifests at time $k+1$ (e.g., treatment at $k=0$ results in the outcome at $k=1$). Leveraging this data, the model aims to provide an unbiased estimate of the ITE $\hat{\delta}(h_t)$ given alternative future treatment sequences $\bar{a},\ \bar{a}'$ prescribed at decision time t.

As shown in Figure 5, the ITE estimation model is first trained with patient data before performing inference. Without loss of generality, the description is provided in terms of a single patient. During training, the sequence of patient data is segmented and used to recursively generate 1-step-ahead predictions. The prediction and the ground truth value are used to compute the loss, which updates the weights of the model. It is important to note that the predicted values could be covariates, treatments, or outcomes, depending on the objective. For instance, while g-computation methods [56] learn to predict covariate evolution, representation balancing methods [9] seek to establish an adversarial game with treatment prediction.

In the inference stage, the data of a testing patient are partitioned based on a selected decision time t. The sequence prior to the decision time constitutes the observed sequence $\bar{X}_t,\ \bar{A}_{t-1},\ \bar{Y}_{t-1}$. From the decision time onward, until the prediction horizon, this observed sequence is used as input to the model to recursively predict the 1-step-ahead outcome given the prescribed treatment a_k at each time step. Finally, the estimate of the time-varying ITE is computed as the difference between the predicted outcomes $\hat{\delta}(h_t) = \hat{Y}^{\bar{a}}_{t+\tau} - \hat{Y}^{\bar{a}'}_{t+\tau}$ given the alternative prescribed treatment sequences $\bar{a} = (a_t,\ \ldots,\ a_{t+\tau-1})$ and $\bar{a}' = (a'_t,\ \ldots,\ a'_{t+\tau-1})$.

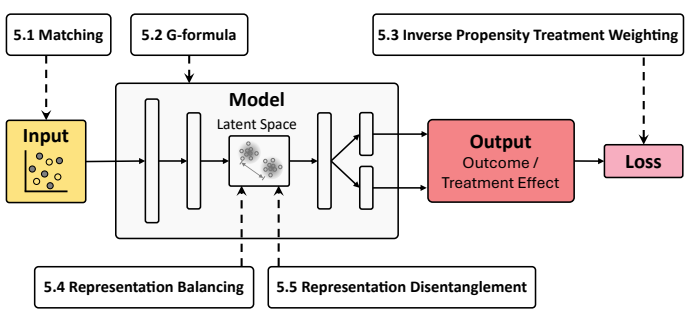

Fig. 6. Figure depicting the locus of adjustment for different adjustment strategies on an exemplary time-varying ITE estimation pipeline. Each adjustment strategy corresponds to a subsection in Section 5.

5 Time-Varying ITE Estimation Methodology

In this section, we review time-varying ITE estimation methodologies. Because the distinction between identification and estimation is often blurred, we organize the literature by how each approach operationalizes identification and adjusts for time-dependent confounding. We begin with methods that fully adopt the sequential identifiability assumptions introduced above (consistency, sequential positivity, and sequential exchangeability) and differ primarily in their adjustment and estimation strategies. We subsequently elaborate on approaches that violate identifiability assumptions, such as sequential exchangeability and sequential positivity. Within each category, we first provide the core intuition behind the adjustment design and subsequently describe the specific methodological frameworks that fall under that category.

As illustrated in Figure 6, adjustment strategies do not operate uniformly; instead, they vary by their *locus of adjustment*, targeting distinct stages of the modeling process: the input space, the model structure, the latent space, or the loss function. This taxonomy reveals a fundamental distinction between classical statistical approaches and modern deep learning adaptations.

Historically, adjustment strategies such as matching (Section 5.1), the g-Formula (Section 5.2), and IPTW (Section 5.3) originated in statistics and epidemiology [57, 81, 82]. These methods typically target the input space, the model structure, or the loss function, respectively. Their primary motivation is to correct for sampling bias externally—modifying the dataset or the optimization landscape to mimic a RCT. With the advent of deep learning, the objective has shifted from external correction to internal constraint. Strategies such as representation balancing (Section 5.4) and disentanglement (Section 5.5) target the latent space, enforcing the neural network to learn a latent representation that is invariant to treatment assignment, thereby removing bias during the feature learning process itself.

Interestingly, deep learning has revitalized the use of classical adjustment strategies. Contemporary architectures enhance traditional estimators by replacing parametric linear models with more flexible function approximators: matching can now be performed by calculating the similarity from latent representations, the g-formula utilizes sequence-to-sequence models to capture complex covariate evolution, and IPTW leverages temporal networks to estimate time-varying propensity scores with greater temporal fidelity.

5.1 Matching

As stated by the fundamental problem of causal inference, it is impossible to observe the counterfactual outcome. To produce reliable estimates of the counterfactual without confounding, matching-based methods build the representation of the counterfactual patient from a cohort of similar patients who are prescribed a different treatment. Modern approaches often employ a similarity metric (or conversely, a distance metric) to weigh or select source units. Examples of such metrics $\text{dist}(\cdot,\cdot)$ between the factual h_f and counterfactual representations h_{cf} include the distance between propensity scores [61], Euclidean distance in the latent space [78], or similarities learned implicitly via attention mechanisms [20]. More formally, the obtained counterfactual representation minimizes the following objective:

$$h_{cf} = \arg\min_{h} \Big(\text{dist}(h_f,\ h) \Big),$$

where f and cf are subscripts for factual and counterfactual, respectively. By enforcing similarity between the factual and counterfactual representations, matching builds an unbiased representation for counterfactual outcome prediction.

The counterfactual representation could be constructed either by identifying the most similar patient or as a weighted sum of the cohort of similar patients (synthetic control). T4 [61] introduces a mini-batch balancing matching strategy in which propensity scores are used as the similarity metric. T4 is first pretrained to estimate the propensity score from the history. Subsequently, patients are sampled to form mini-batches before enhancing them to create pseudo-balanced batches. For every patient in the original batch, the model identifies and includes their most similar neighbors from alternative treatment groups. This process mimics a randomized controlled trial within each outcome prediction training iteration, effectively adjusting for confounding. Another approach, SyncTwin [78], constructs a synthetic twin for each target patient as a weighted sum of the patients in the control group. The weights are obtained by solving the optimization problem that minimizes the difference between the latent representations of the target and the synthetic twin. Drawing insights from synthetic control, SCouT [20] leverages a spatio-temporal Transformer to construct a synthetic representation of the target unit. Unlike static weighting approaches, This framework employs a spatial attention mechanism that dynamically computes the optimal combination of control patients at every time step, thereby implicitly adjusting for time-dependent confounders by altering the synthetic twin's composition as patient states evolve.

5.2 G-formula

The g-formula is a non-parametric identification formula that was first proposed to address time-dependent confounding in longitudinal exposure settings [80]. Despite being originally intended for computing the average causal effect under time-varying treatment strategies, the g-formula can be adapted for the time-varying ITE estimation setting by conditioning on the observed history. Intuitively, the g-formula explicitly models the system's natural dynamics to simulate the counterfactual future. To adjust for future time-dependent confounders within the strata of the observed history $H_t = h_t$, the conditional g-formula employs a "simulate and standardize" mechanism. First, the causal link from future covariates to treatments ($X_k \rightarrow A_k$) is severed by fixing the future treatments to a preset sequence $\overline{A}_{[t,\ t+\tau-1]} = \overline{a}$. Second, the mediated effects ($A_{k-1} \rightarrow X_k$) are preserved by modeling the evolution of future covariates. Finally, the identified estimand is derived by marginalizing (standardizing) over the stochastic evolution of the future covariates. This framework is implemented via g-computation [38, 99], where empirical models (ranging from regression to deep neural networks) are fitted to the observational densities. These models simulate multiple potential trajectories of the future, and the counterfactual outcome is estimated by averaging predictions across these simulated roll-outs, thereby approximating the marginalization over the future covariate distribution.

Under this paradigm, G-Net [56] employs RNN to capture the complex and long-range dependencies in time series data by modeling the conditional distribution of evolving covariates. Instead of predicting all current covariates simultaneously, G-Net factorizes the joint distribution, generating each component of the covariate vector—which explicitly includes the outcome component—sequentially. This generation is conditioned on the learned historical representation and any preceding covariates simulated within the current step. In the context of dynamic regimes, these simulated covariates are combined with the history to determine the intervention for the subsequent time step. Finally, an unbiased estimate of the counterfactual outcome is obtained using Monte Carlo methods that average the simulated values across multiple simulations. Another work [98] adapted G-Net to perform sepsis outcome predictions. In this framework, G-Net is used to simulate the covariate evolution, the simulated covariates are used as input to a separate Long-Short Term Memory classifier for outcome prediction. Instead of using RNNs, the G-Transformer [109] proposes the use of Transformers to capture long-range dependencies. Different from G-Net, G-Transformer factorizes

the covariates into 2 groups: categorical and continuous components. Separate Transformer encoders are used to learn the historical representations of each group in a sequential manner. Building on G-Net or G-Transformer, an approach [21] enables the uncertainty quantification of the estimated outcomes. Leveraging both variational dropout and deep ensembles, this framework captures the uncertainty that originates from the model parameters and better approximates long tailed clinical outcomes. To address the high estimation variance in neural G-computation methods, a different G-Transformer [41] is proposed. Unlike approaches that estimate full covariate distributions via Monte Carlo simulations, this framework reformulates the conditional G-formula as a sequence of nested conditional expectations. The historical representation learned with the Transformer backbone is fed into specific G-computation heads for each future time step. In this framework, the observed outcome at the prediction horizon acts as a ground-truth anchor, allowing the model to recursively optimize preceding predicted outcomes backward through time, ensuring low-variance outcome estimation. CURE [60] implements the outcome model of the g-formula to model point treatments. It pre-trains a domain-specific foundation model using masked language modeling to learn a comprehensive historical representation. The pre-trained model subsequently encodes both the assigned point treatment and patient history into a joint latent representation for outcome prediction. By conditioning on this learned representation, CURE adjusts for static confounders and produces unbiased estimations.

5.3 Inverse Probability of Treatment Weighting

IPTW adjusts for time-dependent confounding by re-weighting the observational data to create a "pseudo-population" where the treatment assignment is independent of the covariates. Individuals receiving treatments that are unlikely given their covariate history are up-weighted, whereas those receiving treatments consistent with the observed confounding patterns are assigned lower relative weights. This re-balancing effectively severs the link from confounder to treatment ($X_k \rightarrow A_k$) while preserving the treatment's mediated effect on the outcome ($A_{k-1} \rightarrow X_k$). Ideally, this removal of bias is represented by *unstabilized weights*, defined as the cumulative inverse probability of receiving the observed treatment sequence:

$$W_{t,\tau} = \prod_{k=t}^{t+\tau-1} \frac{1}{\Pr(A_k = a_k \mid H_k = h_k)}.$$

The conditional probabilities are also known as the propensity scores. However, this can lead to high variance due to the low conditional probability of treatments (extreme weights); practice typically employs *stabilized weights* (*SW*). These introduce a numerator that models the probability of treatment given only the baseline history H_t:

$$SW_{t,\tau} = \prod_{k=t}^{t+\tau-1} \frac{\Pr(A_k = a_k \mid \overline{A}_{[t,\,k-1]} = \overline{a}_{[t,\,k-1]},\ H_t = h_t)}{\Pr(A_k = a_k \mid H_k = h_k)}.$$

In this formulation, the denominator performs adjustment while the numerator reduces the variance of the weights and ensures that the estimator remains conditioned on the specific history H_t of interest.

R-MSN [58] developed a deep learning adaptation of the marginal structural model (MSM) [82] that forecasts outcomes over flexible horizons. The framework consists of 2 components: a propensity network and a prediction network. While the propensity network computes the stabilized weights that account for censoring, the prediction network learns a representation of the observed history to forecast outcomes given alternative treatment sequences. The learned weights are then combined with the prediction error to adjust the overall loss function. To address the challenges of estimating high-dimensional outcomes, a counterfactual generative framework [107] is proposed. The approach

first learns the propensity scores based on the history to calculate the weights. These weights are applied in the loss function to encourage the generator's proxy conditional distribution to resemble the true counterfactual distribution. The generative model trained with this loss function takes a random noise vector and the full treatment sequence as input to generate unbiased, high dimensional counterfactual outcomes. SCIP-Net [40] estimates potential outcomes in continuous time using Neural Controlled Differential Equations (CDEs) and argues that representation balancing (explained in Section 5.4) is not a proper adjustment. This work first reformulates the intractable product integrals necessary for continuous time IPTW adjustment as finite products. This enables the propensity weights and the outcomes to be estimated using separate Neural CDE-based networks.

While the conditional g-formula relies on the correct specification of the outcome distribution conditional on the full history, the IPTW method requires an accurate model of treatment assignment given the time-varying covariates. Unfortunately, the complex underlying DGP and the high dimensionality of time series observational data render the perfect specification of either model challenging. To mitigate the risk of model misspecification and robustly adjust for time-dependent confounders, Doubly Robust (DR) estimators are employed [5]. DR methods combine these approaches by integrating both the outcome and the propensity score models into a unified estimator. This ensures that the overall estimate remains consistent if *either* the outcome or the propensity score model is correctly specified [38, 84].

To mitigate the high variance of doubly robust estimators in low overlap settings, the IVW-DR-learner [25] combines a DR pseudo-outcome with a stabilized weighting scheme. The method employs iterative G-computation to estimate response functions, implicitly marginalizing over future covariate trajectories via nested regressions . These estimates are combined with IPTW to construct a pseudo-outcome based on the efficient influence function, which serves as an unbiased estimator of the counterfactual. To mitigate high variance driven by extreme propensity scores, the final regressor minimizes a loss function weighted by inverse-variance weights (IVW) derived from the variance of the pseudo-outcome.

5.4 Representation Balancing

Representation balancing adjusts for time-dependent confounders by achieving independence between the learned historical representation and the treatment assignment [9, 93]. The essence is that if the latent representation learned from patient history $\Phi(H_k)$ retains information about the treatment A_k (e.g., higher severity correlating with aggressive treatment), confounding bias is present. To sever the links from time-varying covariates to treatments ($X_k \rightarrow A_k$) in the latent space, representation balancing can be formulated as an adversarial game, which can be structured in 2 distinct ways.

The first formulation achieves independence *indirectly* via a minimax game over treatment prediction accuracy [9, 33, 46, 48, 65, 72, 91, 108, 111, 115]. In this formulation, the history encoder (e.g., RNN, Transformer) aims to *maximize* the treatment prediction loss, while the treatment classifier simultaneously aims to *minimize* it. Hence, the objective function can be written as:

$$\min_{\theta_h,\theta_y} \max_{\theta_a} \sum_i \mathcal{L}_y^i(\theta_h, \theta_y) - \lambda \mathcal{L}_a^i(\theta_h, \theta_a)$$

where $\mathcal{L}_y^i$ and $\mathcal{L}_a^i$ denote the outcome and treatment prediction losses for patient data i, respectively. Notations θ_h, θ_y, and θ_a separately represent the learnable parameters of the history encoder, outcome classifier, and treatment classifier. At equilibrium, the treatment classifier performs equivalently to random guessing, suggesting that $\Phi(H_k)$ contains no discriminative information about treatments A_k. The second formulation achieves independence by *directly* minimizing the statistical discrepancy between the latent representations of history across different treatment assignments [1, 23, 63, 105]. This requires the latent representation to be independent of treatment assignment, i.e.,

$$\Pr(\Phi(H_k)) = \Pr(\Phi(H_k) \mid A_k = a_k) \quad \forall a_k \in \mathcal{A}.$$

Different from formulation 1, where the encoder seeks to induce treatment prediction errors, the encoder in formulation 2 aims to minimize the divergence $D(\cdot,\cdot)$ between the joint distribution and the product of the marginals:

$$\min_{\theta_h} \mathrm{D}\Big(\Pr(\Phi(H_k),\ A_k),\ \Pr(\Phi(H_k))\Pr(A_k)\Big).$$

The adversarial component in this formulation functions as a "critic" that assesses the successful alignment between distributions. Both formulations ensure that the historical representation is balanced regarding treatment assignment, effectively adjusting for time-dependent confounders.

In practice, establishing the minimax game under the first formulation can be achieved using a gradient reversal layer (GRL) [9, 33, 46, 48, 91, 108, 111, 115] or by adopting generative adversarial networks (GAN) [72]. Performing gradient reversal is the most common approach to learning balanced representations. Works that adopt this approach generally apply a GRL to the treatment prediction head to construct the minimax game. Designing custom loss functions is an alternative means of implementing adversarial learning. For instance, the Counterfactual Domain Confusion loss [65] promotes the treatment classifier to correctly predict the current treatment while simultaneously minimizing the cross-entropy with a uniform distribution, thus forcing the learned representation to be non-predictive of the treatment assignment. Instead of using the encoder-classifier pair to characterize the adversarial game, GAN is employed to construct a generator-discriminator setup [72]. The generator outputs counterfactuals from the encoded latent representation of history, while the discriminator attempts to discern the factual information.

Constructing the adversarial game under the second formulation can be implemented by simultaneously minimizing a measure of divergence—effectively mimicking the alignment of the joint distribution and the product of the marginals—and the treatment prediction loss. One example would be to minimize both the mutual information (MI) upper bound between the historical representation and the treatment assignment while concurrently minimizing the treatment prediction loss [23]. Minimizing the former aligns the joint distribution of the historical representation and treatment with the marginals, while minimizing the latter promotes accurate treatment classification. Another possible method is through the generator-discriminator architecture. In ACTIN [105], a generator produces historical representations, while a discriminator attempts to distinguish between "real" pairs drawn from the joint distribution and "fake" pairs drawn from the product of the marginals. When equilibrium is reached through adversarial training, independence between historical representations and treatment assignment is achieved. Moreover, it is not only important to ensure that the balanced representation is independent of treatment assignment, but also that the learned representation is invertible. This is to guarantee that no information is lost during encoding; thus, no hidden confounders exist. Causal CPC [23] achieves this by first encoding the past and future histories into separate views. Leveraging the Information Maximization principle, the MI between these views is maximized to guarantee that the context representation is reconstructible. A separate historical representation, which is non-linearly mapped from this context, is then learned to

be treatment-invariant.

Recently, approaches for learning balanced representations have moved beyond constructing complex adversarial games. The work by [63] extends the approach for learning balanced representations in the static setting [93] to the time-varying domain without constructing the adversarial game. The method [93] framed in the static setting established that the counterfactual prediction error is bounded by the sum of the factual error and the discrepancy between treatment and control distributions in the representation space. Building on this idea, the work by [63] identifies that such bounds can be loose. Instead, it introduces Sub-treatment Group Alignment (SGA), which posits that treatment groups are composed of latent sub-groups (e.g., demographic clusters). Instead of forcing global invariance, SGA aligns these corresponding sub-structures individually. By utilizing the Wasserstein-1 distance as a measure of distance between treatment distributions, this work theoretically demonstrates that the counterfactual error is bounded by the weighted sum of these sub-group discrepancies, resulting in a tighter objective. Both Wasserstein distance and maximum mean discrepancy (MMD) fall under the category of integral probability metrics [93]. In the context of survival analysis, TV-SurvCaus [1] employs a hybrid approach, utilizing MMD for representation balancing and IPTW for loss weighting to mitigate time-varying confounding. Given the dynamic treatment sequence and the balanced representation, its survival prediction network outputs probabilities for discretized hazards to estimate the survival function. Parallel to these distance-based methods, Mamba-CDSP [103] proposes parameter-level decorrelation in state space models (SSM) to mitigate the "over-balancing" problem in adversarial representation balancing. The model regularizes the selective parameters of the Mamba architecture to remove the cross-covariance between the hidden historical representation and the current treatment assignment. By mathematically constraining the SSM's internal transition dynamics, it theoretically yields tighter error bounds than standard adversarial balancing approaches.

In addition to properly adjusting for time-dependent confounders to alleviate confounding bias, a variety of additional challenges exist in the problem setup. In clinical scenarios, treatments may be administered concurrently; the value of the prescribed treatment may be continuous; and a future treatment sequence could span a significant duration, where the potential outcome at a long horizon requires estimation. To provide an accurate estimation of the outcome, dedicated modeling considerations are necessary to tackle these challenges.

The occurrence of simultaneous treatments creates mixed causal effects, where individual contributions are obscured by mutual causal interactions. Operating under the assumption that the total causal effect can be decomposed into individual treatment effects and synergies due to mutual interactions, TCFimt [108] disentangles these components using the effect disentanglement block. The model constructs an anchor for individual treatments by combining each specific treatment, its corresponding representation, and the factual outcome. This anchor is utilized within a contrastive InfoNCE loss to maximize the MI with the predicted factual outcome while minimizing the MI with counterfactuals, thereby isolating the treatment synergy effects from individual contributions and enhancing estimation performance.

RCAB [115] addresses the limitations of handling continuous dosages. Establishing the adversarial game requires the adversarial component—the treatment discriminator—to approximate the conditional distribution of treatment given the latent representation of history. However, there is insufficient data to reflect the entire dosage distribution when the treatment has continuous values. To address this, RCAB designs a continuous imbalance loss that leverages neighboring treatment similarity. For a specific treatment, the loss function utilizes weighted aggregation so that samples with similar dosage values contribute to the loss. This compels the generator to produce historical representations that are indistinguishable to the discriminator, even when similar dosage values are considered.

In the τ-step-ahead prediction setting, the number of possible treatment sequences experiences a combinatorial explosion, resulting in ineffective GAN training. To tackle this challenge, EDTS [72] proposes the Sequential Counterfactual Discrimination method. This identifies the counterfactuals that are first-degree relatives of the factual treatment sequence, thus constraining the number of counterfactuals that the treatment and dosage discriminator needs to discern.

The aforementioned methods address the complexity of the *intervention*; yet, they largely operate under the assumption of discrete time steps. However, this is inconsistent with the clinical reality in which time series health data are inherently irregular and obtained continuously. TE-CDE [91] proposes an encoder-decoder architecture built upon Neural CDE backbones. The method operates under the assumption of continuous time sequential randomization, which posits that the observation intensity process is independent of future potential outcomes. To estimate counterfactuals, TE-CDE employs an encoder to learn a continuous latent history, followed by a decoder that evolves this state conditional on hypothetical future treatments. CF-GODE [48] performs counterfactual outcome estimation in multi-agent dynamic systems where agents co-evolve in continuous time. The framework extends the assumption to account for interactions among agents, thereby requiring the potential outcome to be independent of both the treatment assignment and neighbor interference, given the history and graph structure. The model starts by encoding baseline covariates into the baseline latent representation. Subsequently, CF-GODE develops the Treatment-Induced GraphODE approach that models the evolution of the latent representation by combining graph neural networks (GNN) with ODEs. The graph component allows interference from neighboring agents to be aggregated and injects the impact of treatments. Building on this, CAG-ODE [46] further models the effect of simultaneous treatment and evolving interactions. Unlike CF-GODE, which assumes a static graph structure, CAG-ODE utilizes a spatio-temporal GNN encoder to initialize a coupled GraphODE that jointly models the evolution of nodes and dynamic edges. Furthermore, it introduces a treatment fusion module to learn time-dependent representations of concurrent treatments, capturing their combined dynamic effects instead of injecting raw treatment values.

5.5 Representation Disentanglement

Contrary to representation balancing, where confounding information is removed from the general representation of patient history, representation disentanglement decomposes the latent representation of patient history into domain representations. For instance, decomposing into an instrumental, a confounding, and a prognostic/outcome factor [7], i.e.,

$$\phi(H_t) \longrightarrow \{\phi_I(H_t),\ \phi_C(H_t),\ \phi_P(H_t)\}.$$

As the name suggests, the instrumental factor is *only* informative about treatment; the confounding factor is informative of *both* treatment and outcome; and the prognostic factor is *only* informative about the outcome. By structurally separating these components, this mechanism enables "selective adjustment". Confounding bias is removed by balancing or penalizing only the shared factors, while other factors preserve their predictive power.

DCRN [7] aims to mitigate the over-regularization issue in representation balancing by adjusting only for the confounding factor. After decomposing the historical representation into the 3 aforementioned factors, both the confounding and outcome factors are used for outcome prediction. To adjust for the confounding factor, weights are used in a manner similar to IPTW, where propensity scores are derived from the confounding factor. It is equally vital to ensure that the outcome factor is free of confounders. DCRN aligns the treatment distributions using discrepancy measures

such as the MMD. Because the confounding factor is not removed, as in representation balancing, this approach prevents over-regularization during adjustment. Another approach, DR-CRN [17], draws inspiration from multi-task learning to disentangle patient histories into 2 distinct latent factors: a treatment representation and an outcome representation. The model employs separate RNNs to learn these factors—one for outcome prediction and one for treatment assignment classification. To alleviate confounding bias, this method minimizes the MI upper bound between these latent representations, thus enforcing independence and removing confounders from both factors.

5.6 Other Methods

In addition to the aforementioned categories for confounder adjustment, other possibilities have been explored. COSTAR [66] uses self-supervised learning to obtain expressive component representations. It pretrains a Transformer-based encoder to ensure that the learned covariates, treatments, and outcome representations are distinct yet informative. By framing counterfactual estimation as a domain adaptation problem, the model leverages these distinct representations to transfer predictive power across counterfactual domains without traditional balancing constraints. INSITE [51] does not adjust the data space but directly learns the closed-form ODEs that govern the system dynamics due to the imposed treatments. Knowing that individual heterogeneity exists in addition to global laws, this framework first uncovers population-level equations before patient-specific refinements. The population-level equations are uncovered using ODE discovery as a sparse combination of non-linear functions. Subsequently, the coefficients for this equation are refined for each patient by solving an optimization problem. If the discovered ODE accurately captures the true structural equations of the system, the resulting patient-specific equations reflect the underlying causal mechanisms, thereby enabling unconfounded outcome estimation.

Furthermore, there are time-varying ITE works that either accept confounding as an open-challenge and do not perform adjustment [43] or fail to account for confounders rigorously [52, 114]. Rather than introducing a novel confounder adjustment mechanism, BNCDE [43] emphasizes the importance of uncertainty quantification in continuous time treatment effect estimation. The framework employs Neural CDEs to model the continuous latent progression of patient states. Crucially, the time-varying weights of Neural CDEs are modeled by latent Neural Stochastic Differential Equations (SDE). Hence, BNCDE quantifies both epistemic and aleatoric uncertainty: the stochasticity of the Neural SDEs captures model uncertainty, while the prediction head outputs both the predicted outcome and its variance, thereby capturing outcome uncertainty. CDS [114] constructs a 2-stage architecture where the propensity scores are estimated before being used as input to predict survival probability. This model design effectively conditions on a proxy of the covariate but does not account for the mediating effect. [52] posits that patient history can be unraveled into a trend and a seasonality factor, assuming that the former is the source of confounding. While confounding in trend has been accounted for, the theoretical justification for why the seasonality component is confounder-free is omitted.

5.7 Relaxing Sequential Exchangeability

Despite most works assuming strict sequential exchangeability (i.e., no unmeasured confounders), it is difficult to verify the validity of this assumption in reality. For instance, EHRs do not contain information regarding the patient's socioeconomic status, which is, in fact, a critical factor determining the quality of care. Without accounting for the unmeasured confounders, the backdoor path remains open, resulting in biased estimates of the counterfactual outcome. Hence, a series of works relax sequential exchangeability and model potential hidden confounding.

A significant proportion of the work that accounts for hidden confounders explicitly models unobserved multi-cause confounders: confounders that influence multiple treatments when they are prescribed simultaneously. These approaches relax sequential exchangeability to sequential *single* exchangeability, assuming the absence of single-cause confounders. Time Series Deconfounder [8] explicitly learns a latent representation to capture multi-cause hidden confounders. The framework implements a factor model that ensures treatments are conditionally independent at each time step. While the RNN learns the multi-cause confounder representation from patient history, the multi-task predictor estimates the assignment probabilities of all treatments. By inferring these hidden confounders and conditioning on them in a subsequent outcome model, the framework enables unbiased treatment effect estimation. VTDNet [18] learns a latent representation that is not only a proxy for the unobserved multi-cause confounders but also incorporates information for covariate reconstruction and outcome prediction. Employing a Transformer-based variational autoencoder architecture, the model encodes patient history to obtain a Gaussian distribution from which the latent representation is sampled. Akin to the Time Series Deconfounder, VTDNet implicitly assumes a factor model that ensures the conditional independence of treatments if all multi-cause confounders are captured. LipCDE [12] and LipSCDE [11] operate under the assumption that the multi-cause hidden confounders can be decomposed into high- and low frequency components. The low frequency component captures slow varying trends and periodicity, is considered to be approximately time-invariant, and has a bounded influence. The high-frequency component captures fast variations and is assumed to be recoverable through observed covariates acting as noisy proxies. Both works infer a time-varying confounder representation from patient histories and condition outcome models based on this representation. LipCDE uses a Neural CDE to model continuous-time patient state evolution, whereas LipSCDE extends this with a stochastic CDE for outcome prediction.

DSW [62] uses an RNN backbone coupled with attention to infer the latent representation of more general hidden confounders. The representation learning module takes the current treatment, covariates, and the history of previous confounding representations as input to learn a latent variable and update the confounder representation. To ensure that sufficient confounding information is captured, the latent variable obtained from the confounder representation is used to predict both the outcome and treatment. By using the predicted treatment probabilities to obtain the IPTW, the model re-weights the population to produce unconfounded estimates of the individual treatment effects.

While learning a latent representation as a proxy for hidden confounders is a valid approach, DTA [54] considers the observed patient covariates to be a noisy proxy for the true confounders. It uses an RNN-based autoencoder to obtain a latent representation of the hidden confounders from the noisy covariates. To ensure that the latent representation reflects the true hidden confounders, DTA proposes a causal regularization penalty that minimizes the discrepancy between the conditional distributions of the outcomes with and without the information of the current treatment assignment. Minimizing this penalty encourages the treatment assignment to provide no additional information about potential outcomes, conditional on the confounder representation and past treatments, thus approximating sequential exchangeability given the learned hidden confounder representation.

5.8 Relaxing Sequential Positivity

In practice, clinical guidelines and entrenched care conventions make treatment assignment highly non-random: for some clinical states, a treatment is essentially mandated, while for others, it is effectively ruled out, i.e.,

$$\inf_{h_k \in \mathcal{H}_k} \min\left\{\Pr(A_k \mid H_k = h_k),\ 1 - \Pr(A_k \mid H_k = h_k)\right\} = 0.$$

The problem is amplified in time-varying settings, as the covariates are a high-dimensional record of various patient attributes and clinical measurements. As histories become more specific over longer time horizons, the data fragment into increasingly sparse strata. Consequently, for a given detailed history, certain treatment decisions are never (or almost never) observed, causing systematic violations of the sequential positivity assumption.

CF-ODE [19] does not assume sequential positivity in its framework for outcome estimation; rather, it explicitly represents limited overlap as epistemic uncertainty in its counterfactual predictions. The model encodes pre-treatment covariates into a latent state and evolves this state forward under a treatment regime via Neural ODEs, in which treatment enters as an exogenous input. To quantify uncertainty, CF-ODE defines a prior over possible future trajectories and learns a variational posterior parameterized as a Neural SDE. When a specific covariate-treatment pair is rare, the data provide limited constraints on future trajectories, and the variational posterior remains close to the prior. This results in high uncertainty in the final estimate of the outcome.

6 Clinical Application Scenarios

The estimation of time-varying ITE in healthcare is critical for decision-making, as therapeutic strategies must adapt dynamically to a patient's evolving physiological state. Unlike static interventions, optimal patient care is often defined by sequential treatment updates, which require precise decisions regarding three key parameters: treatment type, dosage, and timing. The primary advantage of time-varying ITE estimation models in this context is their ability to handle treatment-covariate feedback loops—where past treatments influence future patient states, which, in turn, influence future treatment decisions. By simulating unobservable counterfactual trajectories, these models allow clinicians to estimate the specific causal effect of a potential intervention sequence before it is administered. To structure this landscape, we categorize applications by their distinct temporal and systemic demands, ranging from the high-frequency volatility of critical care to the longitudinal disease progression of secondary care, and finally to the interdependent network dynamics of public health.

Critical Care. In the high-stakes environment of the ICU, the primary challenge is the volatility of patient stability and the narrow window for effective intervention. While traditional early warning scores (e.g., SOFA, MEWS) are reactive—flagging deterioration only after vital signs have deviated—they are fundamentally predictive, not prescriptive. They indicate that a patient is deteriorating, but not how to intervene. A reliable time-varying ITE model bridges this gap by simulating future states under different treatment plans, effectively moving clinical care from reactive management to proactive prevention. Sepsis, for example, requires navigating complex tradeoffs where individualized care is essential. Recent methodologies utilizing critical care datasets [49, 50] demonstrate the potential to dynamically refine clinical decisions according to the patient's evolving state. In the context of antibiotic stewardship, these models assist clinicians in determining the optimal treatment and timing, balancing mortality reduction against risks such as toxicity accumulation and drug resistance while avoiding specific contraindications like drug-induced kidney injury [98, 106]. Similarly, for hemodynamic management, they facilitate granular decisions regarding fluid resuscitation versus vasopressor use, allowing clinicians to predict adverse events like pulmonary edema under liberal versus conservative strategies [54, 109]. Beyond outcome prediction, these approaches enhance clinical trust by decomposing the influence of specific covariates on treatment response. This interpretability facilitates personalized healthcare by stratifying patients into high- and low-responder subgroups [18], ensuring that aggressive interventions are targeted strictly at those who will derive a causal benefit.

Secondary Care. Moving beyond the rapid physiological fluctuations of the ICU, time-varying ITE estimation is equally critical for managing the long-term, evolutionary progression of chronic diseases inherent to secondary care. In oncology, tumors develop drug resistance over time, altering the treatment effect at different stages of progression [102]. A static model might correctly identify a chemotherapy agent as effective initially, but it fails to detect the inflection point at which the causal benefit diminishes due to acquired resistance or where cumulative toxicity outweighs efficacy. Time-varying ITE estimation models allow clinicians to determine the optimal treatment stopping time by continuously estimating counterfactual trajectories, thus preventing over-treatment when the effect plateaus. To validate these approaches, studies [9, 56, 58, 104] frequently utilize the pharmacokinetic-pharmacodynamic (PK-PD) synthetic dataset [27] to simulate tumor volume trajectories in response to different sequences of chemotherapy and radiotherapy, demonstrating how iterative estimation can identify schedules that optimally balance tumor reduction with patient tolerance. In neurodegenerative disorders such as Alzheimer's Disease (AD) and Friedreich's Ataxia (FA), the clinical priority shifts from shrinking a mass to altering the slope of irreversible functional decline. A critical challenge in this domain is isolating the subtle, time-dependent benefits of therapy from the noisy background of natural progression. By employing the NACC [6] and FA-COMS [79] datasets, time-varying ITE estimation methods demonstrate significant potential for this clinical scenario [18, 20]. These methods not only provide accurate treatment effect estimations that are deconfounded from the natural cognitive progression but also enable the generation of synthetic counterfactuals in rare disease cohorts with small donor pools and decouple the effects of the simultaneously applied medications.

Public Health. Finally, the necessity for time-varying ITE estimation scales from the individual physiology of the patient to the collective epidemiology of public health. In this domain, the clinical objective shifts from reversing organ failure or tumor growth to alleviating viral transmission in the community. Just as a clinician must adapt drug dosages to a patient's evolving vital signs, public health officials must continuously adjust the intensity of policies—such as social distancing or vaccination mandates—in response to changing infection rates [33, 46, 48]. Recent applications [33] demonstrate that these models can estimate the effective reproduction number under dynamic conditions, proving beneficial not only for the retrospective assessment of policy effectiveness but also for prospective planning to recommend optimal strategies for emerging pandemics. Furthermore, these approaches address the critical challenge of network interference, where interventions are not isolated to a single unit but spill over to affect connected populations [46, 48]. For instance, mobility restrictions or vaccination mandates in one region causally mitigate transmission risks in neighboring areas. By capturing this continuous co-evolution of disease propagation and simultaneous interventions, time-varying ITE models empower policymakers to answer complex "what-if" questions regarding the optimal timing and sequencing of public health mandates.

It is imperative to understand that while the aforementioned approaches are validated on datasets that span a broad spectrum of clinical domains, they are not designed to account for the specific demands of the corresponding domains. Nonetheless, time-varying ITE estimation retains significant clinical potential and warrants further investigation.

7 Evaluation

This section illustrates the necessary components for empirically evaluating time-varying ITE estimation models, including the data, evaluation metrics, and the predominantly used baselines for comparison.

7.1 Data

The evaluation of time-varying ITE estimation primarily focuses on factual and counterfactual outcome prediction. While factual evaluation is performed by comparing the predicted values with the ground truth, the fundamental problem of causal inference renders the evaluation of counterfactual predictions on real-world datasets impossible. Consequently, the existing literature relies on synthetic/semi-synthetic datasets for counterfactual prediction evaluation and uses real-world datasets for factual prediction evaluation. The most commonly used datasets can be found in Table 3.

7.1.1 Synthetic Datasets. The PK-PD [27] and CVSim [36] datasets are the predominantly used synthetic datasets that model biological processes using known differential equations. The equations underlying the PK-PD dataset simulate the treatment of Non-Small Cell Lung Cancer, where the tumor cell number changes in response to chemotherapy and radiation. Building on the mathematical model behind the PK-PD dataset, R-MSN [58] first adapted this simulator for causal inference benchmarking by introducing a biased treatment assignment policy. In this adapted dataset, the variables are defined as follows:

- Covariates: Tumor volume history and randomly determined patient subgroups.
- Treatments: Binary applications of fixed-dose chemotherapy and radiotherapy (4 choices).
- Outcomes: The continuous tumor volume.

Crucially, R-MSN incorporates time-dependent confounding by modeling the probability of treatment assignment as a sigmoid function of the patient's tumor diameter history; this mechanism ensures that more severe patients with larger tumors are more likely to receive treatment, thereby creating a realistic observational bias that is controlled by a tunable parameter. In comparison, CVSim simulates the cardiovascular system as an electrical circuit using a lumped-parameter model. G-Net [56] extended the CVSim-6C simulator by adding stochastic elements and treatments; the resulting S-CVSim simulator serves as the foundation for the evaluation dataset. The variables in the CVSim dataset are defined as:

- Covariates: 18 hemodynamic features, including heart rate, pressures, flows, and volumes.
- Treatments: Continuous dosages of fluids and vasopressor applications.
- Outcomes: A subset of continuous and discrete patient covariates, such as mean arterial pressure, central venous pressure, and pulmonary edema.

G-Net infuses time-dependent confounders by designing observational regimes. The probability of assigning a treatment is explicitly calculated as a function of covariates.

7.1.2 Semi-Synthetic Datasets. Fully synthetic datasets provide accessible counterfactual ground truths but suffer from a distributional shift, as the dynamics of differential equations fail to capture the complexity of real clinical data. To bridge this gap, the literature utilizes semi-synthetic datasets that combine real-world covariates with simulated causal mechanisms. In these setups, time-varying patient features are extracted from observational datasets (e.g., MIMIC-III [50], FA-COMS [79], EEG [112], and COVID-19 US [107]) to preserve the complex structure of patient health variations. To establish a ground truth for evaluation, researchers discard the observed treatments and outcomes, instead overlaying a simulated probabilistic treatment policy (to induce controlled time-dependent confounding) and generating synthetic outcomes via a known response mapping. Using the widely adopted MIMIC-III dataset as an example, we describe the simplified semi-synthetic dataset construction steps below [40, 65, 103]. Notations $f(\cdot)$, $g(\cdot)$, and $h(\cdot)$ denote arbitrary functions; k and j separately denote the time and treatment choice indices.

(1) **Extract Covariates** X_k: Real-world time series health data are used to ensure that the complexity of the feature space is realistic. Vital signs and patient demographics are extracted from MIMIC-III to serve as time-varying and static covariates, respectively.

(2) **Simulate Untreated Outcomes** Z_k: Establish a baseline health trajectory. The untreated outcome is generated by combining a longitudinal trend (the patient's endogenous progression over time) with exogenous fluctuations caused by current physiological states (time-varying covariates), plus noise ϵ_k.

$$Z_k = \underbrace{f(k)}_{\text{trend}} + g(X_k) + \epsilon_k$$

(3) **Simulate Treatment Vector** A_k: We introduce time-dependent confounding by modeling the conditional distribution of each treatment component $A_{k,j}$ based on patient history. This creates a biased assignment mechanism where the treatment intensity or probability depends on past severity.

$$\theta_{k,j} = \sigma\Big(\underbrace{h(\bar{Y}_{k-1}, X_k)}_{\text{history representation}} \Big)$$

where $\sigma(\cdot)$ denotes the logistic sigmoid function, mapping the history representation to the probability interval $(0,\ 1)$. The realization $a_{k,j}$ is sampled from a distribution specific to the domain $\mathcal{A}_j$. For binary treatments(i.e., $\mathcal{A}_j = \{0, 1\}$): $\theta_{k,j}$ represents Bernoulli probabilities and $a_{k,j} \sim \text{Bernoulli}(\theta_{k,j})$.

(4) **Calculate the Cumulative Treatment Effect** $E(k)$: Treatments are modeled to have a delayed and decaying impact (e.g., inverse-square decay) rather than merely an immediate influence. The influence is scaled by the assignment probability to correlate treatment efficacy with patient severity.

$$E(k) = \sum_{u \le k} \lambda(k-u) \cdot s\Big(a_u,\ \theta_u\Big)$$

where $\lambda(\cdot)$ is a decay function (e.g., inverse-square), and $s(\cdot)$ computes the treatment influence at time u given the realized treatment vector a_u and the probability vector $\theta_u = (\theta_{u,1},\ \ldots,\ \theta_{u,d_a})$.

(5) **Simulate Treated Outcomes** Y_k: Combine the untreated outcome with the cumulative treatment effects to obtain the treated outcome.

$$Y_k = Z_k + E(k)$$

7.1.3 Real-World Datasets. Complementing these synthetic and semi-synthetic datasets, real-world datasets are employed to evaluate factual prediction performance—the model's ability to accurately predict observed outcomes given the observed history. Although these datasets lack counterfactual ground truth, they provide a critical evaluation of the model's capacity to accurately capture complex clinical patterns. The MIMIC-III database [50] remains the standard for this purpose in critical care. However, recent literature has expanded validation to diverse clinical environments, such as neurodegenerative disorders using NACC [6] and cardiovascular diseases using CPRD [39]. Furthermore, studies focusing on learning the interactions between agents in the system have utilized infectious disease datasets, specifically COVID-19 US [107] and COVID-19 Germany [97]. A typical formulation of the MIMIC-III dataset for factual outcome evaluation is presented below [8, 23, 35, 103].

- Covariates: Vital signs and lab tests constitute the time-varying covariates; patient demographics, such as gender and age, constitute static covariates.
- Treatments: Binary applications of vasopressors and mechanical ventilation (4 choices).

- Outcomes: Continuous-valued diastolic blood pressure.

Table 3. Clinical datasets used for time varying ITE estimation evaluation. The name, evaluation type (synthetic, semi-synthetic, or real-world), specific clinical domain, examples of covariates, examples of treatments, examples of outcomes, the raw/original dataset sample size, and the range of sample sizes used for experimentation in existing works. N/A stands for not applicable, and "-" represents unspecified.

Dataset	Type	Clinical Domain	Covariate Example	Treatment Example	Outcome Example	Raw[a] Size	Range[a]
PK-PD [27]	Synth.	NSCLC	Tumor volume history	Chemo, RT	Tumor volume	N/A	1200–21,000
CVSim [36]	Synth.	Cardiovascular	Hemodynamic variables	Fluids, Vaso	MAP, PO	N/A	11,702–13,000
Cardiovascular [59, 110]	Synth.	Cardiovascular	Hemodynamic variables	Fluid intake	ABP	N/A	-
MIMIC-II [90]	Real	Nephrology	Creatinine history	Dialysis	Creatinine	~800	231–428
MIMIC-III [50]	Real	ICU	Vital signs, Lab tests	Vaso, MV	DBP	46,467	2,313–11,715
MIMIC-III [50]	Semi	ICU	Vital signs, Lab tests	-	-	46,467	1,000–11,715
MIMIC-IV [49]	Real	ICU[b]	Vital signs, Lab tests	Fluids	Fluid overload	180,733	3,553–8,934
AUMCdb [100]	Real	Sepsis	Vital signs, Lab tests	Antibiotics	SOFA score	20,109	3,136–6,285
CAMP [31]	Real	Asthma	Lung function metrics	No treatment	PreFF	1,041	275
FA-COMS [79]	Real	FA	Quality of life scales	Calcitriol	FARSn, 9-HPT, ADL	410	25
NACC [6]	Real	AD	Vital signs, Biomarkers	Statins, NSAIDs	AD Risk	3,410[b]	31,059
EEG [112]	Semi	Alcoholism	Electrode values	-	-	122[b]	1,000[c]
COVID-19 US [107]	Real	COVID	Deaths, Symptoms	Mask mandate	New cases	3,219[c]	3,219[c]
COVID-19 DE [97]	Real	COVID	-	-	Active cases	401[c]	401[c]
COVID-19 EU [33]	Real	COVID	COVID tests	Stay-at-home orders	Effective reproduction number	7[c]	7[c]
CPRD [39]	Real	CVD	HDL Cholesterol, Creatinine	Statins	LDL	~6.9%[b]	125,784
MS CCAE [67]	Real	Stroke	Medical history	Aspirin, Rivaroxaban	Stroke	$>10^8$	2,955,399

[a] The default units are #simulations for synthetic dataset and #patients for semi-synthetic and real-world datasets.
[b] Dataset Clarification: The MIMIC-IV dataset contains both ICU admission data and general hospital admissions; the NACC database contains 3,410 patients at its initial launch in 2007, but the data have been accumulating over the years; the EEG datast contains 122 patients, each with 120 trials; the CPRD dataset has raw size ~6.9% UK population.
[c] unit Clarification: EEG dataset uses #trials; COVID-19 US uses #counties; COVID-19 DE uses districts; COVID-19 EU uses countries.

7.2 Evaluation Metrics

The performance measures of time-varying ITE models are stratified by the availability of ground truth and the specific component of the causal inference pipeline being assessed. While the ultimate objective is accurate treatment effect estimation—which necessitates learning the unbiased causal effect of the prescribed treatments—the literature employs a composite evaluation strategy. This includes outcome metrics to validate the model's approximation of the observed

patient trajectories; oracle-based metrics, which utilize a known data-generating process (the "oracle") to access true potential outcomes and unobservable counterfactuals, to validate treatment effect estimation and treatment prediction capabilities on (semi-)synthetic datasets; and representation metrics to quantify/visualize de-confounding.

7.2.1 Outcome Prediction. The prerequisite for accurate causal inference is the ability to learn a reliable representation of the patient's history and outcome mechanism. Outcome prediction compares the model's predicted outcome $\hat{y}_i$ against the ground truth outcome y_i. These metrics are applicable to both real-world datasets for evaluating factual outcomes and (semi-)synthetic datasets for evaluating counterfactual outcomes. For continuous outcomes, such as tumor volume in the PK-PD dataset or diastolic blood pressure in MIMIC-III, the **Root Mean Squared Error (RMSE)** is the standard metric for quantifying the deviation of the predicted outcome from the observed reality [9, 58, 65, 103]. For a population of size N, this is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{4}$$

In scenarios where outcomes are discrete or binary—such as the occurrence of stroke or pulmonary edema—existing works rely on threshold-independent metrics to evaluate discriminative capability [60, 98, 109]. The **Area Under the Receiver Operating Characteristic Curve (AUROC)** measures the trade-off between the true positive rate (TPR) and the false positive rate (FPR):

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t))dt \tag{5}$$

Additionally, the **Area Under the Precision-Recall Curve (AUPRC)** is standard for imbalanced data, focusing on the trade-off between precision and recall for the positive class:

$$\text{AUPRC} = \int_0^1 \text{Precision}(\text{TPR}^{-1}(t))\ dt \tag{6}$$

7.2.2 Treatment Effect Estimation. To validate that the model has successfully disentangled the causal effect from confounding bias, metrics must compare the estimated ITE against the ground truth value. This is exclusively performed on synthetic and semi-synthetic datasets using the **Precision in Estimation of Heterogeneous Effect (PEHE)** [19, 44, 61, 62]. It measures the MSE between the true ITE, $\delta(h_{i,t})$, and the predicted ITE, $\hat{\delta}(h_{i,t})$:

$$\text{PEHE} = \frac{1}{N}\sum_{i=1}^{N}(\delta(h_{i,t}) - \hat{\delta}(h_{i,t}))^2. \tag{7}$$

It is worth noting that variations of the PEHE metric have been developed to address specific reporting needs and data limitations. For example, IF-PEHE [3, 60] is introduced to calculate the error between the estimated treatment effects and the approximated true effects derived via influence functions that *do not* require oracle access. Minimizing PEHE confirms that the model achieves valid counterfactual reasoning, accurately predicting outcomes under alternative treatment sequences to obtain the ITE estimation.

7.2.3 Treatment Prediction. While a low RMSE indicates predictive precision, it does not strictly guarantee clinical utility. For example, a model could achieve a low RMSE in outcome prediction but still fail to correctly rank the effectiveness of different treatment plans. To assess decision-support capability, **Accuracy** is utilized to evaluate the selection of discrete treatment types and optimal administration timing [9, 91, 108]. Notably, because counterfactual

outcomes are unobserved in real-world scenarios, this metric is evaluated using synthetic datasets where an oracle provides ground truth outcomes for all potential treatment sequences. The "optimal" treatment a_i^* is defined as the specific intervention plan that maximizes the patient's health utility (e.g., minimizing tumor volume) at a future horizon. Mathematically, the accuracy of selecting the optimal treatment type or timing is defined as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\hat{a}_i^* = a_i^*\right), \tag{8}$$

where $\mathbb{I}(\cdot)$ is the indicator function, and $\hat{a}_i^*$ and a_i^* represent the predicted and true optimal treatment type/timing, respectively.

For treatment timing, the evaluation treats the temporal dimension as a discrete set of intervention choices. The model predicts the potential future outcome $Y_{t+\tau}$ that would result from initiating treatment at each possible delay δ (where $\delta \in \{0, \ldots, \tau - 1\}$). The predicted optimal timing $\hat{a}^*$ is the specific start time that leads to the best final outcome:

$$\hat{a}^* = \underset{\delta \in \{0,\, \ldots,\, \tau-1\}}{\operatorname{argmin}} \; \hat{Y}_{t+\tau}\left(\overline{a}_{[t+\delta,\; t+\tau-1]}\right). \tag{9}$$

Accuracy is then calculated as the frequency with which $\hat{a}^*$ matches the ground truth optimal timing a^*. High accuracy in this context indicates the model's capacity to identify the most effective treatment strategy, thus translating predictive estimates into actionable clinical recommendations.

7.2.4 Representation Evaluation. As shown in Table 2, a substantial proportion of approaches leverage representation balancing for confounder adjustment. This requires learning a "balanced" latent representation $\Phi(H_t)$, where the distribution of patient history is invariant across treatment groups. To quantify the quality of balancing, **t-SNE** [64] is frequently used [9, 46, 48, 105] to visualize the latent space. In a successfully balanced representation, the treated and control samples should overlap significantly (indicating de-confounding), rather than forming distinct, separable clusters based on treatment assignment. Another representation disentanglement method also resorted to t-SNE for qualitative assessment [17]. Similarly, the outcome component of the appropriately disentangled representation should also demonstrate significant overlap and is thus not predictive of treatments.

7.3 Baselines

To rigorously assess the contribution of novel time-varying ITE architectures, it is vital to benchmark performance against established methods that represent distinct modeling paradigms. The predominantly used baselines can be categorized into classical statistical methods, discrete-time deep learning models, and continuous-time deep learning models.

Classical Statistical Baseline. **MSM** [82] is the preferred classical statistical baseline in the literature. Despite their linearity, MSMs remain the requisite statistical baseline; they provide crucial verification that the complexity of deep learning models yields statistically significant benefits over traditional methods. It is worth mentioning that while linear regression functions as a baseline for confounded performance, the necessity of adjustment for counterfactual outcome prediction has been repeatedly validated [9, 58, 62].

Discrete-time Deep Learning Baseline. For discrete-time deep learning comparisons, existing works predominantly utilize **R-MSN** [58], **G-Net** [56], **CRN** [9], and **CT** [65]. These are selected because they represent distinct approaches to de-confounding: R-MSN implements recurrent network-based IPTW adjustment; G-Net employs the g-formula

with covariate evolution modeling; and CRN and CT perform representation balancing adjustment by establishing an adversarial game. Additionally, CT also provides a baseline for a model's capacity to handle long-range temporal dependencies.

Continuous-Time Baselines. A critical distinction in modern evaluation is the handling of irregular sampling and continuous dynamics. Consequently, **TE-CDE** [91] has emerged as the standard baseline for continuous-time ITE estimation evaluation. Unlike discrete-time models that require data discretization or imputation, TE-CDE operates under the assumption of an underlying continuous latent path. It is therefore the necessary comparator for demonstrating that a proposed model correctly handles the irregular nature of real-world clinical events.

Emerging Standards for Evaluation. While the aforementioned baselines provide a robust foundation for comparative analysis, the rapid evolution of causal architecture necessitates an expanded scope for rigorous evaluation. Recent literature has introduced diverse modeling strategies that challenge previous benchmarks: **IVW-TR-learner** [25] adapts metalearning frameworks to the time-varying ITE estimation domain; and **ACTIN** [23] advances representation balancing through a generator-discriminator framework. In the continuous-time domain, innovation has moved beyond standard Neural CDEs, with **SCIP-Net** [40] coupling IPTW with Neural CDEs to strictly enforce de-confounding, and **INSITE** [51] performing explicit ODE discovery for interpretable estimation. Consequently, to ensure evaluation reliability, it is essential for modern benchmarking to account for these emerging frameworks, thereby confirming that new methods are "State-of-the-Art" approaches that indicate a generalized improvement in causal inference capability.

8 Current Challenges

Despite significant methodological advancements in time-varying ITE estimation, the adoption of these approaches from theoretical formulation to clinical deployment is hindered by substantial difficulties. This section critically examines the primary challenges facing the field. We first address the fragility of the identifiability assumptions in real-world healthcare settings. We subsequently analyze the complexities introduced by the intrinsic properties of observational data, specifically irregularity and continuous-time dynamics. Finally, we discuss the crisis of evaluation—the inability to validate models against ground truth and the subsequent reliance on imperfect simulations.

8.1 Assumption Violations

Causal inference methods rely on a set of identifiability assumptions that are mathematically necessary but often unrealistic in clinical practice. The violation of these assumptions compromises the validity of the estimated treatment effects.

Violation of Sequential Positivity. The sequential positivity assumption posits that every patient has a non-zero probability of receiving each treatment option at every time step. In healthcare, clinical guidelines often force deterministic decision-making, leading to distinct violations of this assumption. For example, in patients with septic shock, the probability of not receiving antibiotics approaches zero [92]. Moreover, the presence of absolute contraindications indicates that the prescription of certain treatments will be harmful to the patient, rendering a violation that persists regardless of the sample size [77]. Patients with acute kidney injury are almost never subjected to NSAIDS because NSAIDs inhibit prostaglandins, which are essential for maintaining blood flow in the kidneys [32]. In these regions of the covariate space, counterfactual estimation relies entirely on extrapolation from dissimilar patients, leading to high-variance estimates and unreliable policy recommendations. While few methods have addressed violations of

sequential positivity in observational data [19, 42, 73], most works rely on this assumption to ensure identifiability for causal effect estimation.

Violation of Sequential Exchangeability. Most time-varying ITE models assume that there are no unobserved confounders conditional on the patient history. The hidden confounders influence both the treatment assignment and the outcome, resulting in biased estimation of the causal effect. However, this assumption is difficult to satisfy in observational healthcare data. Clinicians make treatment decisions based on a holistic view of the patient, including visual cues (e.g., patient behavior) and verbal exchanges that are rarely captured in the data. Additionally, factors such as the Social Determinants of Health (SDOH)—including financial instability, housing, and transportation access—are potent drivers of both treatment assignments and outcomes; yet, they are systematically missing from standard clinical datasets [14, 29]. To mitigate this, existing frameworks generally resort to learning a latent representation of the hidden confounders from high dimensional patient history data, operating on the premise that hidden confounders induce learnable correlations among observed data [8, 35, 62]. However, confounders such as SDOH factors often retain a weak "footprint" in the observed data, rendering them virtually undiscoverable. A significant proportion of the deconfounding literature addresses multi-cause confounders—factors that influence multiple treatments [8, 11, 12, 18]. This assumes the absence of single-cause confounders, such as specific logistical constraints that are mathematically indistinguishable from the direct treatment effect and cannot be inferred from the data [83]. Moreover, even if the multi-cause confounders are recovered in the latent representation, it is possible for the representation to inadvertently capture mediators and colliders, which will introduce additional biases for estimation [74].

Violation of Consistency. The consistency assumption requires that the observed outcome for a treated individual is the same as the potential outcome had they been assigned that treatment. This assumes that the "treatment" is well-defined without multiple versions. In reality, medical treatments are heterogeneous. While pharmaceutical variations (e.g., dosage) are often standardized or explicitly recorded, procedural treatments such as surgery contain vast, unmeasured variability in surgeon skill, operative technique, and duration. When multiple versions of a treatment exist but are aggregated into a single binary variable (e.g., "Surgery: Yes/No"), the consistency assumption is violated, and the estimated effect becomes an ill-defined average weighted by the specific distribution of versions in the historical data. Importantly, estimating the treatment effect under multiple versions of treatment is different from considering each version as a separate treatment [101]. More specifically, instead of asking "what is the causal effect of surgery A version A?", the question is "what is the effect of surgery A given the natural distribution of treatment versions inherent to the version assignment mechanism?". This distinction is crucial when specific versions cannot be distinguished for the entire population, but it requires adjusting for specific confounders that affect both the treatment and the version assignment.

8.2 Challenges From Data

While violations of the identifiability assumption pose a fundamental threat to the validity of estimation, the practical application of time-varying ITE estimation is further constrained by the intrinsic nature of time series healthcare data. Unlike the regularly sampled time series found in macroeconomics [34], healthcare data is a byproduct of clinical workflows, resulting in complex temporal structures that render standard discrete-time approximations inadequate [13].

Continuous Time Dynamics. Physiological processes and the effects of medical treatments operate in continuous time; patient health states evolve dynamically instead of as static snapshots. However, standard ITE frameworks typically

employ discrete-time architectures, where discretization introduces a fundamental mismatch between the model and the underlying biological mechanism known as temporal aggregation bias [16]. By aggregating data, binning diminishes the precise temporal ordering necessary to estimate causal effects. For instance, in an ICU setting, a vasopressor administered at 10:15 am may cause a rise in blood pressure at 10:20 am. If these events are binned into a single hourly-window, the cause and effect appear simultaneous, and the causal effect is unidentifiable.

To circumvent the limitations of discretization, recent frameworks leverage Neural ODEs/CDEs/SDEs to model the evolution of the patient state continuously [43, 46, 48, 91]. These methods offer a significant theoretical advantage: by defining the hidden state dynamics via a parameterized derivative, they naturally accommodate irregular time intervals and allow for the estimation of treatment effects at any arbitrary point in time. Unfortunately, the applicability of these models to physiological data is constrained by the strict requirement for Lipschitz continuity. To guarantee numerical stability and a unique solution, these frameworks implicitly enforce a bounded rate of change [15]. This smoothness constraint contradicts stiff physiological systems—processes characterized by long periods of stability interrupted by sudden, rapid phase transitions, such as the precipitous drop in blood pressure during septic shock [53, 95]. Consequently, while modeling with neural differential equations solves the discretization problem, their mathematical constraints risk obscuring the critical turning points that define high-stakes clinical events.

Irregularities. Observational healthcare data is characterized by high sparsity and irregularity. However, the challenge is not merely technical (i.e., missing data handling) but fundamentally causal: the missingness mechanism is informative, which violates the standard Missing At Random assumptions [88, 96]. In clinical practice, the sampling mechanism—the decision to observe a variable—is often correlated with the patient's underlying state. A patient in critical condition will have clinical measurements recorded frequently, while a stable patient will be monitored sparsely. Consequently, the sampling intensity manifests as a proxy for the hidden confounder that simultaneously affects both the treatment and the outcome [2].

Effectively integrating this irregular signal remains difficult for current deep learning architectures. Existing works typically employ Last Observation Carried Forward (LOCF) imputation to satisfy the input requirements of deep learning architectures [40, 65]. Yet, LOCF modifies the causal information contained in the observation process. By deterministically copying previous values forward, it conflates the lack of measurement with clinical stability, making uniformly valued segments ambiguous between true stability and prolonged non-observation. This attenuates the sampling intensity signal and simultaneously induces structured, time-varying measurement error, the magnitude of which is correlated with the latent patient state and past treatments. Hence, the proxy signal that could be leveraged to learn hidden time-dependent confounding is diminished while introducing additional biases to the data.

8.3 Evaluation Challenges

In addition to the challenges caused by the tenuous assumptions and the irregular, continuous-time nature of healthcare data, evaluating time-varying ITE frameworks remains difficult for two reasons. First, only the factual outcomes are accessible, while the counterfactual outcomes are inherently unobservable, eliminating direct ground truth for individual-level effect accuracy. Second, limited reproducibility, which stems from incomplete public access to code and preprocessing pipelines—makes it difficult to independently verify results and compare methods consistently across studies.

Unobservable Counterfactuals. The primary challenge in evaluation is the reliance on proxy ground truth. As explained in Section 7.1, the standard approach utilizes synthetic or semi-synthetic datasets, where counterfactual outcomes are

generated via known functions. These synthetic ground truths enable ITE estimation error metrics (e.g., PEHE) to be computed; allow sensitivity analysis to be conducted across varying confounding strengths and degrees of overlap; and provide experimental reproducibility, whereby multiple realizations of a known DGP can be used for evaluation. However, synthetic evaluation is an imperfect substitute for clinical validity [28]. This is because simulators are often domain-specific and depend on compartmental simplifications and structural rigidity [27, 36]. This fundamentally misrepresents the stochasticity and complexity of real physiological processes, where multi-system interactions, drug resistance, and additional unknown factors defy closed-form representations. Furthermore, these simulations fail to capture the informative sampling and irregular missingness in time series healthcare data. As a result, models that excel on these benchmarks may simply be overfitting to the inductive biases of differential equations rather than learning robust causal representations [10, 75].

Reproducibility. Advancements in digital healthcare are heavily dependent on the transparency and reproducibility of reported methods. In time-varying ITE estimation specifically, a meaningful proportion of published methods summarize their implementation but are not accompanied by end-to-end code releases. Among works that do release code, it is common for reproducibility to be limited to synthetic datasets, while the procedures for conducting real-world experiments are not fully captured. As time-varying ITE estimation hinges on subtle identifiability and data-generating assumptions, offering a complete and reproducible implementation is equally important to developing the underlying theoretical foundation; it is a prerequisite for reliable scientific accumulation.

9 Future Directions

Following the summarization and analysis of the boundaries of existing research, we propose three critical frontiers for future investigation: the integration of survival analysis for time-to-event outcomes, the leveraging of multi-modal data to mitigate hidden confounding, and the development of Digital Twins for robust evaluation.

9.1 Time-to-Event Outcomes

Current time-varying ITE research predominantly focuses on continuous scalar outcomes (e.g., blood pressure, tumor size) or binary classifications (e.g., mortality at a fixed horizon). However, in many clinical scenarios, the quantity of interest is not just *if* an event occurs, but *when* it occurs. For instance, if a binary mortality outcome is used to measure the effectiveness of a drug for aggressive glioblastoma, all patients may eventually die within that window and the drug will be deemed ineffective. However, time-to-event analysis reveals the true clinical benefit by demonstrating that the drug extends median survival time, capturing the crucial value of delayed progression. This necessitates the integration of survival analysis, which models the time duration until an event of interest occurs.

Unlike the standard regression setup, time-to-event analysis must account for *censoring*—instances where the observation window ends or a patient is lost to follow-up before the event occurs. This complexity is compounded by *competing risks*, where the occurrence of one event (e.g., death) precludes the occurrence of another (e.g., hospital discharge). Future research must extend counterfactual estimation to model the hazard function $\lambda^a(t)$ that jointly represents the instantaneous rate of occurrence for the competing event under the prescribed treatments. By combining deep survival models with ITE estimation, researchers can derive patient-specific survival curves and compute the difference in survival probability or mean survival time under alternative treatment sequences.

9.2 Multi-Modality

The prevailing dependency on structured tabular data (e.g., vital signs and lab tests) in current time-varying ITE estimation methods creates a fundamental validity challenge, notwithstanding their significant strides in modeling temporal dynamics. Since clinical decision-making is intrinsically multimodal—often driven by visual cues in medical imaging or nuanced observations in unstructured clinical notes—models developed solely with tabular data inevitably fail to capture the complete patient history, resulting in a structural violation of the sequential exchangeability assumption. To address this, future efforts should be devoted to the development of multi-modal representation learning frameworks capable of mapping heterogeneous data sources into a unified latent space for the ensuing adjustment and treatment effect estimation. Realizing this potential presents substantial technical challenges beyond multi-modal fusion. From a general modeling perspective, architectures must account for data sparsity and asynchronous temporal alignment, as high-dimensional modalities such as MRI or pathology reports are acquired far less frequently than bedside vitals. In the causal context, additional problems are introduced regarding valid representation; the multi-modal data exacerbates the curse of dimensionality, resulting in violations of the sequential positivity assumption [22]. Furthermore, as discussed in Section 8.2, the presence of informative sampling must be considered to preserve the integrity of the identifiability assumptions. By incorporating the full spectrum of information available to the clinician, multi-modal frameworks simultaneously reduce confounding in data and learn a more comprehensive representation of the patient, thereby minimizing confounding bias and enabling reliable time-varying ITE estimation in complex healthcare settings.

9.3 Digital Twin

A Digital Twin (DT) in healthcare is characterized as a high-fidelity, dynamic virtual replica of a physical patient that continuously updates its internal state with the real-world individual. In the context of ITE estimation, a DT can function as a solution to causal transportability. Transportability refers to the ability to transfer causal information learned from a source population to a target domain that may differ in its underlying mechanisms. Current ITE models typically estimate a CATE based on population statistics. A Digital Twin advances this by serving as a personalized generative simulation engine that begins by capturing global physiological dynamics and treatment responses across a large cohort. It subsequently finetunes to an individual's continuous data stream to update structural priors, effectively narrowing the causal reference class to a single unit. By continuously aligning its internal state with the physical patient, the DT facilitates the generation of parallel counterfactual trajectories under differing treatment sequences. This capability directly resolves the fundamental evaluation challenge by providing a synthetic, yet physiologically grounded, ground truth for "what-if" analysis.

However, realizing Digital Twins for ITE requires significant future research efforts. This includes focusing on hybrid architectures that integrate mechanistic physiological models with deep learning to ensure that generated counterfactuals remain biologically plausible outside the distribution of the training data. Furthermore, to support high-stakes clinical decision-making, these models must provide rigorous uncertainty quantification and address the computational complexity inherent in simulating continuous-time dynamics for real-time deployment.

10 Conclusion

This literature review serves to bridge the gap between the theoretical advancements in time-varying ITE estimation and practical applications in clinical settings. Through a comprehensive examination of theoretical foundations, deep learning-based methodologies, evaluation frameworks, and current clinical applications, we have mapped the trajectory

of the field while delineating key challenges and future research directions. This highlights a fundamental paradigm shift: moving from static, population-level protocols toward dynamic, individualized treatment planning. As deep learning architectures become increasingly capable of disentangling the complex, non-linear relationships inherent in modern healthcare data, they empower the generation of reliable causal estimations—marking a definitive step closer to the realization of truly personalized healthcare.

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A Notations

Table 4. Summary of notation for the time-varying ITE setup.

Symbol	Type	Domain	Description
Indices and Spaces			
k	Index	$\mathbb{N}_0$	Discrete time step identifier
t	Index	$\mathbb{N}_0$	Decision time index (current time)
τ	Constant	$\mathbb{N}_{\geq 1}$	Estimation horizon (time steps into the future)
$d_x,\ d_a$	Constant	$\mathbb{N}_{\geq 1}$	Dimensions of covariate and treatment spaces
$\mathcal{X}$	Set	$\mathbb{R}^{d_x}$	Space of observed covariates
$\mathcal{A}$	Set	$\mathbb{R}^{d_a}$	Space of treatments (continuous or discrete)
$\mathcal{Y}$	Set	$\mathbb{R}$ or $\{0, 1\}$	Space of outcomes
Observed Variables (Random & Realizations)			
X_k	Random Vector	$\mathcal{X}$	Covariates at time k
x_k	Realization	$\mathcal{X}$	Observed covariate values at time k
$\overline{X}_t$	Random Sequence	$\mathcal{X}^{t+1}$	History of covariates $(X_0,\ \ldots,\ X_t)$
$\overline{x}_t$	Realization	$\mathcal{X}^{t+1}$	Observed sequence of covariates
A_k	Random Vector	$\mathcal{A}$	Treatment assignment at time k
a_k	Realization	$\mathcal{A}$	Observed treatment value at time k
$\overline{A}_{t-1}$	Random Sequence	$\mathcal{A}^t$	History of past treatments $(A_0,\ \ldots,\ A_{t-1})$
$\overline{a}_{t-1}$	Realization	$\mathcal{A}^t$	Observed sequence of past treatments
H_t	Random Vector	$\mathcal{H}_t$	Full history vector $H_t = (\overline{X}_t,\ \overline{A}_{t-1})$
h_t	Realization	$\mathcal{H}_t$	Observed full history value at t
$Y_{t+\tau}$	Random Variable	$\mathcal{Y}$	Factual outcome observed at time $t + \tau$
$y_{t+\tau}$	Realization	$\mathcal{Y}$	Observed value of the factual outcome
Potential Outcomes and Inference			
$\overline{a} = \overline{a}_{[t:t+\tau-1]}$	Fixed Sequence	$\mathcal{A}^\tau$	Prescribed future treatment sequence (deterministic), alternative sequence denoted as $\overline{a}'$
$\overline{a}_{[0:t+\tau-1]}$	Fixed Sequence	$\mathcal{A}^{t+\tau}$	Full sequence of treatments including past observations and future prescriptions $(\overline{a}_{t-1}, \overline{a}_{[t:t+\tau-1]})$
$Y^{\overline{a}}_{t+\tau}$	Random Variable	$\mathcal{Y}$	Potential outcome at $t + \tau$ under prescribed treatment sequence $\overline{a}$
$\hat{Y}^{\overline{a}}_{t+\tau}$	Estimate	$\mathcal{Y}$	Predicted outcome value produced by the model under the prescribed treatment sequence $\overline{a}$
$\delta(h_t)$	Scalar	$\mathbb{R}$	Estimand for ITE/CATE defined as $\mathbb{E}[Y_{t+\tau}(\overline{a}) - Y_{t+\tau}(\overline{a}') \mid H_t = h_t]$
$\mathrm{ATE}_{t,\tau}(\overline{a},\ \overline{a}')$	Scalar	$\mathbb{R}$	Estimand for ATE defined as $\mathbb{E}[Y_{t+\tau}(\overline{a}) - Y_{t+\tau}(\overline{a}')]$